A factorized Pauli–Villars multigrid method for the Möbius domain-wall operator

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The Möbius domain-wall operator has a spectrum that surrounds the origin, which defeats non-Hermitian Krylov solvers and most multigrid constructions. Kanamori, Chen and Matsufuru showed that preconditioning with the Pauli–Villars operator together with the site-diagonal part of the operator, $A' = (D_{\mathrm{PV}}C_{\mathrm{PV}}^{-1})^{\dagger}DC^{-1}$, moves the spectrum into the right half-plane and gives the smallest condition number among the operators they examined. This note describes a two-grid method for A' on the even sites. Its coarse-grid correction is factorized into one Galerkin solve for the quark operator and one for the adjoint Pauli–Villars operator, each with a nearest-neighbor coarse stencil, so that the outer system can be the even-odd one while the coarse operators stay nearest-neighbor. The prolongation is built from approximate low modes lifted to the whole lattice, and the correction is exact on the vectors it is made from. The smoother is a fixed Chebyshev polynomial. Why the method converges is not obvious: the field of values of the preconditioned operator reaches far into the left half-plane, so the usual convergence bound of GCR does not apply. Its eigenvalues nevertheless stay at $\mathrm{Re}\,\lambda \geq 0.78$, and its pseudospectra on 12^4 and 16^4 bound the convergence of full GMRES; the truncated GCR used in practice converges best when the coarse solves are inexact, and with accurate ones it can stagnate. On a quenched $24^3 \times 48$ configuration the multigrid iterations grow by a factor of 2–4 from $m = 0.1$ to $m = 0.001$, against 20 for even-odd CGNE, and at $m = 0.001$ the solve is 7.5–9.0 times faster than CGNE in double precision and 4.5–4.8 times faster than in mixed precision, the setup being paid back after 2–5 right-hand sides; a whole point-source propagator at $m = 0.001$ takes half the time of the mixed-precision CGNE, setup included. The results are from one configuration per volume; Sec. B describes how to extend them.

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I. Introduction

Lattice computations of flavor physics and of the hadronic contributions to precision observables need a fermion discretization whose chiral symmetry is broken only mildly at finite lattice spacing. The overlap operator [1, 2] has an exact lattice chiral symmetry. Domain-wall fermions [3, 4, 5] approximate it at a lower cost by adding a fifth dimension of extent L_s, in which modes of opposite chirality are bound

to the two boundaries by a large negative Wilson mass $-M_5$; the residual chiral symmetry breaking is suppressed as L_s grows, and the effective four-dimensional operator is a rational approximation of the overlap operator. The Möbius generalization [6, 7, 8] introduces two coefficients (b, c) and improves this approximation at fixed L_s; $c = 0$ is Shamir's operator.

The price is paid in the linear solver. The operator is neither Hermitian nor positive, one application costs L_s four-dimensional Wilson hops, and the standard solver is still the conjugate gradient method on the normal equation of the even-odd preconditioned operator, whose condition number is the square of that of the operator itself. For Wilson-type fermions this problem has been solved by multigrid: the low modes that slow down Krylov solvers are locally coherent [10], so a few of them, cut into blocks, span the low-mode space, and adaptive aggregation multigrid makes the cost nearly independent of the quark mass [11, 12, 13].

Multigrid for domain-wall fermions. The domain-wall operator has a spectrum that encloses the origin, because of the large negative Wilson mass. This violates the half-plane condition of non-Hermitian Krylov methods, whose convergence is then of the order of the system size [15, 16]. Boyle [16] lists four multigrid approaches that have had a measure of success, and they differ in the operator that is coarsened. Cohen *et al.* [17] coarsened the squared unpreconditioned operator, with blocks covering the whole fifth dimension, and noted that its coarse operator reaches two hops. HDCG [18] accelerates the best standard solver, CG on the normal equation of the red-black (even-odd) operator, and coarsens that squared Schur complement; its coarse stencil is next-to-next-to-next-to-nearest neighbor, 81 points for blocks of at least 4^4. HDCR [19] works instead directly on the nearest-neighbor Hermitian indefinite operator $\gamma_5 R_5 D$ at $c = 0$, so that the coarse operator stays nearest neighbor, explicitly to make re-evaluation along an HMC trajectory affordable. Brower, Clark, Howarth and Weinberg [21] pointed out that the Pauli–Villars preconditioned operator $D_{\mathrm{PV}}^{-1} D$, not D, carries the physical four-dimensional degrees of freedom, solved the system $D_{\mathrm{PV}}^{\dagger} D x = D_{\mathrm{PV}}^{\dagger} b$ with coarse operators built from the projected Wilson kernel, and demonstrated it in the two-dimensional Schwinger model. Boyle and Yamaguchi [20] compared these approaches for Shamir fermions; HDCR could be set up and solved faster than a single red-black CG solve on large volumes. Two of their observations bear on the present method: among red-black solvers, CG on the Schur complement $M_{ee} - M_{eo} M_{oo}^{-1} M_{oe}$ needed far fewer iterations than on the left-preconditioned $1 - M_{ee}^{-1} M_{eo} M_{oo}^{-1} M_{oe}$; and combining a red-black smoother with the HDCR coarse correction was not effective, which they attributed to the eigenvectors of the red-black operator and of $\gamma_5 R_5 D$ not being aligned.

Recent work by Boyle and collaborators. The multiple right-hand-side HDCG of Ref. [16] works on the red-black Hermitian operator $(M_{ee} - M_{eo} M_{oo}^{-1} M_{oe})^{\dagger} (M_{ee} - M_{eo} M_{oo}^{-1} M_{oe})$ with a flexible ADEF2 two-level conjugate gradient, a coarse space spanned by blocked low modes obtained with Chebyshev filters, and a stationary coarse solver (deflation plus a Chebyshev polynomial). Treating several right-hand sides at once with block CG and batched matrix products makes the coarse space sub-dominant, and gains of up to a factor of twenty over red-black CG are reported at physical quark mass. In a Lattice 2026 talk, Boyle and Oare [22] named two limitations of this approach—the cost of forming the 81-point coarse operator, and that a single solve on a new gauge field, as in HMC, does not profit from many right-hand sides—and presented a multigrid that directly coarsens the two-hop operator $D_{\mathrm{PV}}^{\dagger} D$ with blocks that contain the whole fifth dimension, which gives a 33-point coarse stencil. They emphasized that $D_{\mathrm{PV}}^{\dagger} D$ is non-normal: its field of values can extend into the left half-plane although its spectrum does not, the bound of Eisenstat, Elman and Schultz [23] for GCR is then lost, and they introduced a shift in the smoothers. The coarse solve dominated the cost; a three-level multiple right-hand-side version performs similarly to multiple right-hand-side HDCG, and an exact dense inverse on a further coarse level is planned.

Setup cost and ensemble generation. The setup of a multigrid, and of deflation, is paid once per gauge field, which is affordable for propagators but not for the solves of Hybrid Monte Carlo, where the field changes at every step. Eick and Mawhinney [34] avoid the setup by renormalization-group blocking: a fine

Möbius ensemble ($a^{-1} = 2\,\mathrm{GeV}$) is blocked to a coarse lattice ($a^{-1} = 1\,\mathrm{GeV}$) whose native domain-wall operator approximates the low modes of the fine one, and the coarse solve corrects a CG on the normal equations. They find that the interpolated correction carries high modes that the fine operator amplifies by four orders of magnitude, and that a Chebyshev low-pass filter of degree 150–300 is needed to make the correction useful; they also find the fifth-dimensional profiles of the fine and coarse low modes to differ.

Site-diagonal and Pauli–Villars preconditioning. Kanamori, Chen and Matsufuru [24] examined the spectra of several preconditioned Möbius operators. Combining the Pauli–Villars factor with the site-diagonal part C of the operator, they found that $A' = (D_{\mathrm{PV}} C_{\mathrm{PV}}^{-1})^{\dagger} D C^{-1}$ has eigenvalues with positive real part and the smallest condition number among the candidates, while replacing D_{PV} by a block-diagonal truncation brings back eigenvalues with negative real part. Independently, Neff's modification of the domain-wall matrix [25, 26] rescales the fifth dimension by a site-local factor Q and changes the conditioning of the even-odd system.

This note. We describe a two-grid method for A'. It is an adaptive aggregation-based multigrid in the sense of Refs. [11, 12, 13]: the coarse space is built from computed near-null vectors, cut into aggregates, and the coarse operators are Galerkin projections of fine operators, not a rediscretization; it differs from these methods in that the coarse-grid correction is factorized rather than the Galerkin correction of the operator that is solved. The outer system is the usual even-odd reduced one, as in HDCG, but the coarse-grid correction is *factorized*: the inverses of the quark operator and of the adjoint Pauli–Villars operator are replaced by two separate Galerkin inverses, each with a nearest-neighbor coarse stencil and an aggregation that is compatible with the five-dimensional chirality. The prolongation is built from approximate low modes lifted to the whole lattice in two ways, which makes the correction exact on these vectors for the right form of the operator. The smoother is a fixed Chebyshev polynomial and the setup combines Chebyshev filtering with a bootstrap. The note states the method and gives first numerical results: the field of values, spectrum and pseudospectra of the preconditioned operator, the behavior of truncated GCR, a mass scan and point-source propagators against even-odd CGNE, and the sensitivity to the parameters.

Section II describes the method in words. Section III recalls the spectra on which the construction rests. Sections IV–X describe the method: the operator and its Pauli–Villars preconditioning, its even-odd forms, the factorized correction and the prolongation, the coarse solvers, the setup, the smoother and the outer solver. Section XII gives the results: first why the method converges, then its cost against CGNE. Section XIII lists the open problems. Appendix A relates the notation to that of other references, and Appendix B sketches the implementation and how to reproduce and extend the results.

Conventions. Five-dimensional fields are $\psi_s(x)$, $s = 0, \ldots, L_s - 1$; $P_+, P_- = \frac{1}{2}(1 \pm \gamma_5)$. R reverses the fifth dimension, $(R\psi)_s = \psi_{L_s-1-s}$, and $\Gamma_5 = \gamma_5 R$. Subscripts e and o denote blocks on the even and odd four-dimensional sites; X_{pq} maps parity q to parity p. The injection ι puts a field on the even sites onto the whole lattice (zero on the odd sites), and $\pi = \iota^{\dagger}$ takes the even sites. The operator and its parts C and Q are defined in Sec. IV; Appendix A relates them to the notation of Refs. [8, 24] and of Bridge++ [14].

II. Overview of the method

This section describes the method in words, for readers who know domain-wall fermions but not necessarily multigrid; the sections after it give the equations.

Multigrid in one paragraph. A multigrid preconditioner removes the error of an iterate in two parts. The part made of high modes is damped by a few steps of a cheap iteration on the fine lattice, the *smoother*. The part made of low modes, the near-zero modes that make CG slow, is removed by solving a much smaller problem on a *coarse lattice*. The coarse lattice comes from *aggregation*: the fine lattice is cut into blocks (here 4^4 sites with the whole fifth dimension), each block becomes one coarse site, and on each

block a few approximate low modes, the *test vectors*, are kept as its degrees of freedom. The *prolongation* $\mathcal{P}$ maps a coarse vector to the fine lattice by these vectors and $\mathcal{P}^\dagger$ maps back; the coarse operator is the *Galerkin* projection $\mathcal{P}^\dagger X \mathcal{P}$ of a fine operator X, which couples only neighboring coarse sites when X couples only neighboring fine sites. A few vectors a block suffice because low modes are locally coherent [10]. Finding the test vectors is the *setup*, paid once per gauge field and shared by all right-hand sides.

What is solved. As for even-odd CGNE, $D\psi = \eta$ is reduced to the even sites, here with the Pauli–Villars and site-diagonal preconditioning of A': a system $\tilde{A}y = \rho$ in the left form $\tilde{A} = \tilde{A}_L$, which can carry Neff's α (Sec. V); ψ is reconstructed from y afterwards. The outer solver is flexible GCR keeping two directions, GCR(2), with one two-grid cycle as the preconditioner in each iteration (Sec. X).

One cycle. A cycle is a Chebyshev polynomial in $\tilde{A}$ (the smoother), a coarse correction, and the smoother again (Algorithm 3). The coarse correction is the new ingredient. Coarsening $\tilde{A}$ itself, a four-hop operator on the even sites, would give a coarse stencil of 81 points; instead its two factors are inverted separately, first the adjoint Pauli–Villars operator and then the quark operator, each through the Galerkin operator of D_{PV} or D, whose coarse stencil has 9 points (Fig. 4, Sec. VI). Both coarse systems are solved only roughly, to a relative residual of about 0.1, in single precision and on the even coarse sites (Sec. VII).

The setup. (i) A Chebyshev filter applied to random vectors gives approximate low modes y of $\tilde{A}_R$, the vectors of its smallest singular values; (ii) a bootstrap improves them: the cycle is built from the current vectors, and a few FGMRES steps preconditioned by it, applied to each vector, enrich it in what the cycle still misses; (iii) each y is lifted to the whole lattice in two ways, s and $\hat{t}$, one for each coarse solve, which makes the correction exact on these vectors (Proposition 1); (iv) the lifts are split by their Γ_5-chirality and orthonormalized on each block, which gives $\mathcal{P}$, and the coarse operators of D and D_{PV} are built from it (Algorithm 2, Sec. VIII). One setup serves both forms and every α.

Where to read on. The construction is in Secs. V and VI; Table 1 compares it with other multigrid methods for domain-wall fermions. The parameters used in practice are in Table 4, and the comparison with CGNE in Sec. XII.F.

III. Spectra of preconditioned Möbius operators

The location of the spectrum decides which iterative methods can work. A Krylov residual polynomial p with $p(0) = 1$ that is small on the spectrum exists at low degree only if the spectrum can be enclosed by a region, such as an ellipse, that excludes the origin. The same holds for the Chebyshev smoother of Sec. IX, which is defined by an ellipse in the right half-plane. In short, Krylov solvers and polynomial smoothers work when the eigenvalues lie on one side of the origin, and those of the domain-wall operator do not; the preconditioners below move them there. The spectra of Ref. [24], reproduced in Figs. 1–3, show what the preconditioners do. The lattices are given in the captions.

The unpreconditioned operator. The free spectrum (Fig. 1) is a band that reaches into $\mathrm{Re}\,\lambda < 0$. On a $2 + 1$-flavor configuration at the physical point (Fig. 2, top left) the real parts lie in $[-1.29, 13.2]$. Conjugate gradients and BiCGStab therefore do not apply to D itself, and the practical choice has been CG on a positive definite squared operator.

Site-diagonal preconditioning. With the site-diagonal part C, whose inverse is a constant matrix in the fifth dimension and spin (Sec. IV.C), DC^{-1} has real parts in $[-0.22, 2.22]$ (middle left): the extent shrinks, the sign problem remains.

Pauli–Villars preconditioning. The operator $D_{\mathrm{PV}}^{-1}D$, with $D_{\mathrm{PV}} = D(m = 1)$, is a rational approximation of the overlap operator [21]. Because D_{PV}^{-1} is as expensive as a solve with D, Ref. [21] used $D_{\mathrm{PV}}^\dagger$ instead. $D_{\mathrm{PV}}^\dagger D$ has positive real parts (top right), and its low modes lie close to the imaginary axis with a positive

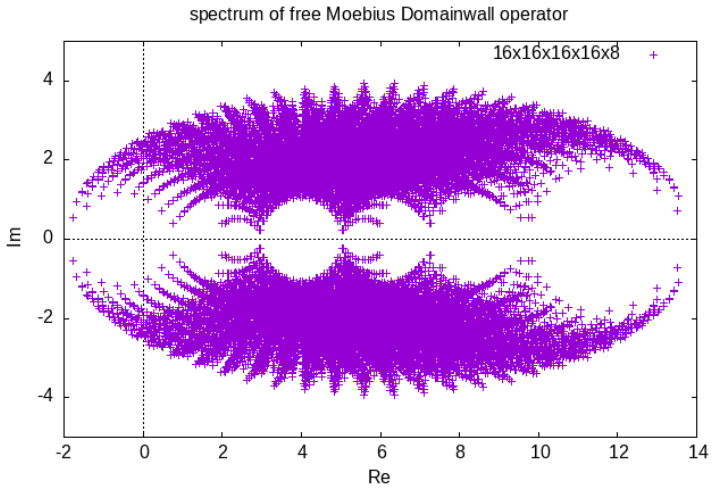


FIG. 1. Spectrum of a free Möbius domain-wall operator on $16^4 \times 8$ with quark mass 0.01, $b = 1.5$, $c = 0.5$ and domain-wall height $M_5 = 1$. Reproduced from Ref. [24], Fig. 1.

shift of the order of the quark mass, as expected of an approximate overlap operator; its high end, however, reaches about 187, which an overlap operator would not show. Replacing D_{PV} by its block-diagonal approximation B_{PV}, which drops the hops between 4^4 blocks, brings back negative real parts (middle and bottom right).

Both together. The operator $A' = (D_{\mathrm{PV}} C_{\mathrm{PV}}^{-1})^\dagger D C^{-1}$ has real parts in $[2.67 \times 10^{-4},\ 5.22]$ (Fig. 3, left); its condition number, 1.96×10^4, is smaller than the 2.34×10^4 of $D_{\mathrm{PV}}^\dagger D$, and the smallest among the operators examined in Ref. [24]. With D_{PV} replaced by B_{PV} the real parts become negative again (right). A preconditioner that truncates the Pauli–Villars factor to blocks would therefore inherit eigenvalues with negative real part; in the method below the factor is kept exact on the fine lattice and is approximated only through its Galerkin coarse operator (Sec. VI).

Two caveats. First, the figures show the full-lattice operator A'; the even-odd operators of Sec. V are related to it but their spectra are not part of these figures. Second, positivity of the real parts of the eigenvalues is not positivity of the field of values. For a non-normal operator the convergence bound of GCR [23] requires the Hermitian part $(A + A^\dagger)/2$ to be positive definite, which can fail while every eigenvalue lies in the right half-plane; this is the mechanism discussed for $D_{\mathrm{PV}}^\dagger D$ in Ref. [22]. It occurs for the even-odd form: for $\tilde{A}_R$ itself the field of values reaches just below zero on 12^4 ($\mathrm{Re}\, W \geq -4.6 \times 10^{-3}$ with stout smearing, -5.0×10^{-2} without), and for the multigrid-preconditioned $\tilde{A}_R \mathcal{M}$ far below (Sec. XII.B). The bound of Ref. [23] is then lost, but the spectrum of $\tilde{A}_R \mathcal{M}$ stays at $\mathrm{Re}\, \lambda \geq 0.78$ and its pseudospectra bound full GMRES (Secs. XII.C, XII.D); truncated GCR is not covered (Sec. XII.E). For the full-lattice A' it has not been measured.

IV. The operator and Pauli–Villars preconditioning

This section defines the Möbius operator in the form of Brower, Neff and Orginos [8] and derives from it the quantities the method is built on: the part C of the operator that acts within a four-dimensional site and its inverse, the Γ_5-transformation of D, the Pauli–Villars preconditioned operator A' of Ref. [24], and Neff's α. Every step is written out. The notation follows Ref. [8] except where stated; Appendix A relates it to Ref. [24] and to Bridge++ symbol by symbol.

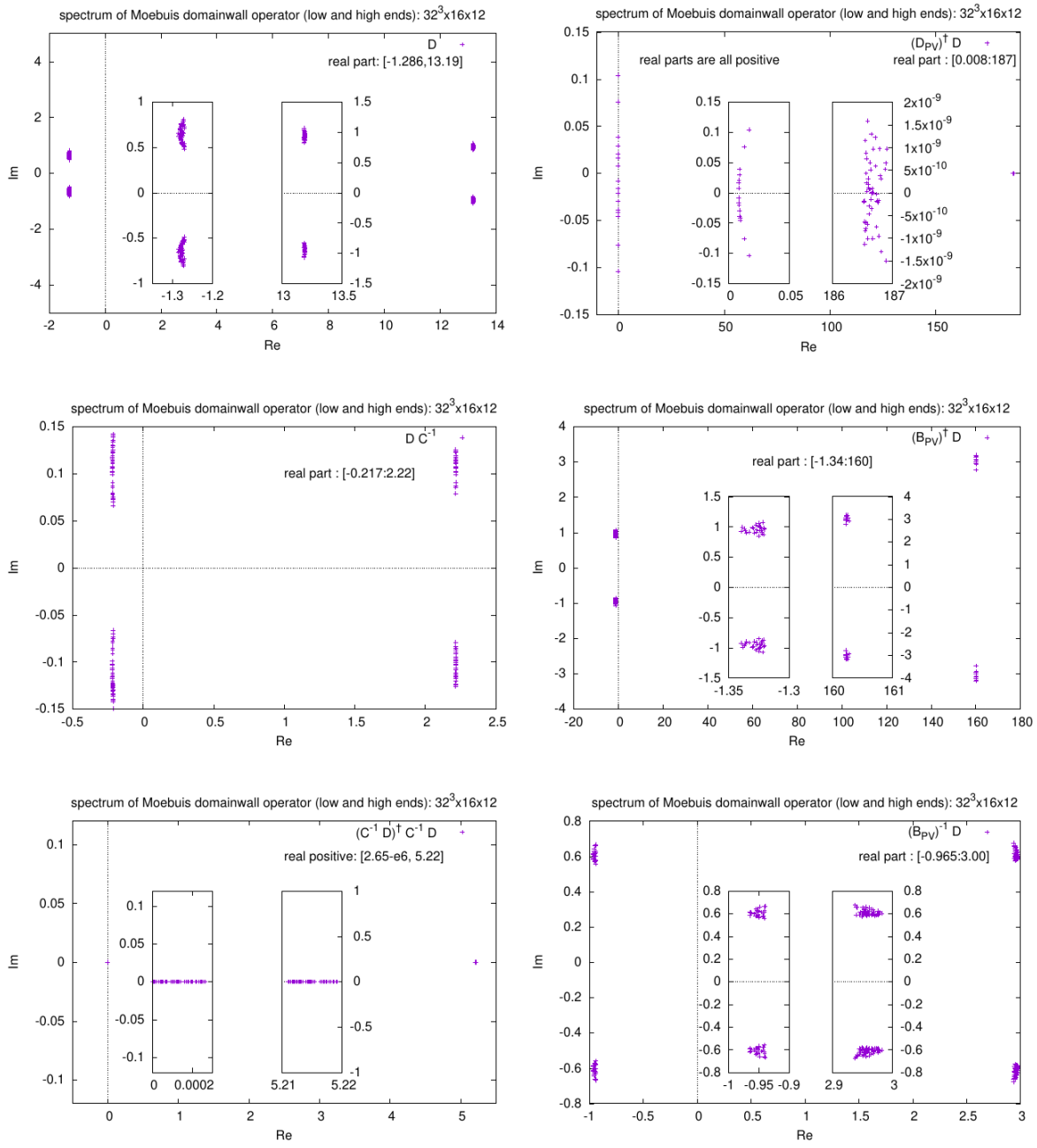


FIG. 2. Low and high ends of the spectrum of the Möbius operator on a JLQCD $2+1$-flavor configuration at the physical point, with preconditioners. Left column: D, DC^{-1}, $(C^{-1}D)^\dagger C^{-1}D$. Right column: $D_{\rm PV}^\dagger D$, $B_{\rm PV}^\dagger D$, $B_{\rm PV}^{-1}D$, where $B_{\rm PV}$ is the block-diagonal approximation of $D_{\rm PV}$ (inverted approximately in the last panel). The text of Ref. [24] quotes $48^3 \times 16$ with $L_s = 12$; the panel titles read $32^3 \times 16 \times 12$. Reproduced from Ref. [24], Fig. 3.

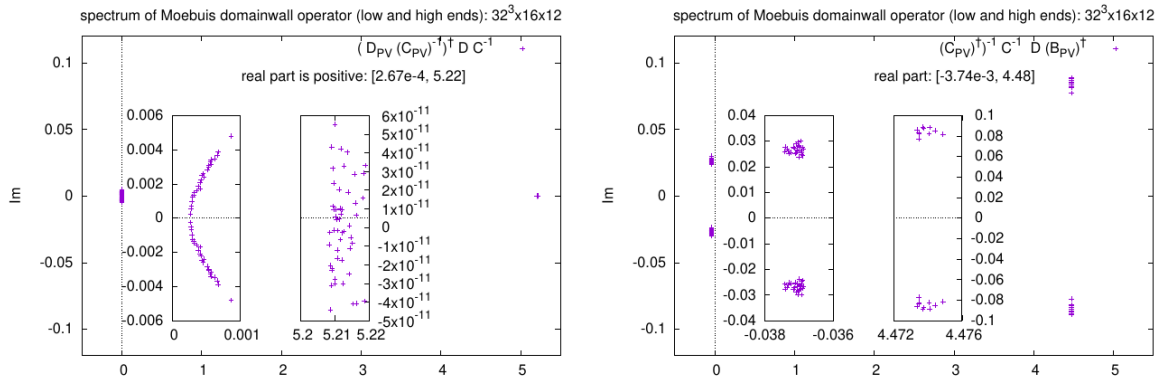


FIG. 3. Left: spectrum of $A' = (D_{\rm PV}C_{\rm PV}^{-1})^\dagger D C^{-1}$, Eq. (4.18), on the configuration of Fig. 2; all real parts are positive. Right: the same with $D_{\rm PV}$ replaced by its block-diagonal approximation $B_{\rm PV}$; negative real parts reappear. Lattice size as in Fig. 2 (the text of the source quotes $48^3 \times 16$, $L_s = 12$; the panel titles read $32^3 \times 16 \times 12$). Reproduced from Ref. [24], Fig. 4.

A. The Möbius operator

Ingredients. ψ is a field on the four-dimensional lattice with an extra index $s = 0, \ldots, L_s - 1$, the fifth dimension; ψ_s is an ordinary Dirac field for each s. $P_\pm = \frac{1}{2}(1 \pm \gamma_5)$ are the chiral projectors. The Wilson hopping term is

$$(K\psi)(x) = \sum_{\mu=1}^{4} \Big[(1-\gamma_\mu) U_\mu(x) \psi(x+\hat{\mu}) + (1+\gamma_\mu) U_\mu^\dagger(x-\hat{\mu}) \psi(x-\hat{\mu}) \Big], \tag{4.1}$$

and the Wilson operator with mass $-M_5$ (the domain-wall height $M_5 > 0$) is

$$W = (4 - M_5) - \tfrac{1}{2} K. \tag{4.2}$$

This is the usual $D_W(-M_5)$ with hopping coefficient $\frac{1}{2}$; in the κ normalization $W = (4-M_5)(1-\kappa K)$ with $\kappa = 1/(8-2M_5)$. W acts on each slice s in the same way and does not mix slices. Its free spectrum, needed in Sec. XI, is $\{-M_5 + \sum_\mu (1 - \cos p_\mu) \pm i(\sum_\mu \sin^2 p_\mu)^{1/2}\}$.

The operator. With

$$D_+ = 1 + bW, \qquad D_- = cW - 1 \tag{4.3}$$

(Ref. [8], Eq. (2.18)), the Möbius operator at quark mass m acts slice by slice as

$$(D\psi)_s = D_+ \psi_s + D_- \big(P_- \psi_{s+1} + P_+ \psi_{s-1} \big), \qquad \psi_{L_s} \equiv -m\,\psi_0, \quad \psi_{-1} \equiv -m\,\psi_{L_s-1} \tag{4.4}$$

(Ref. [8], Eq. (2.20), with s counted from 0). For $L_s = 4$, as a matrix in s,

$$D = \begin{pmatrix} D_+ & D_-P_- & 0 & -mD_-P_+ \\ D_-P_+ & D_+ & D_-P_- & 0 \\ 0 & D_-P_+ & D_+ & D_-P_- \\ -mD_-P_- & 0 & D_-P_+ & D_+ \end{pmatrix}. \tag{4.5}$$

Shamir's operator is $b = 1$, $c = 0$; the results of this note use $b = 1.5$, $c = 0.5$ ($b - c = 1$, $b + c = 2$). The quark mass enters only through the two corners. The physical four-dimensional quark field is $q = P_-\psi_0 + P_+\psi_{L_s-1}$.

The hop in the fifth dimension. Eq. (4.4) separates into a part that hops in the fifth dimension and one that does not. Define

$$(Y\psi)_s = -P_-\psi_{s+1} - P_+\psi_{s-1} \qquad \text{(with the same wrap-around)}, \tag{4.6}$$

the off-diagonal part of Eq. (4.5) without D_- and with a minus sign. Then $D = D_+ - D_-Y$. The order matters: D_- stands to the *left* of Y, since in Eq. (4.4) D_- acts on the shifted and projected field. Inserting Eqs. (4.2) and (4.3),

$$D = 1 + bW - (cW-1)Y = 1 + Y + W\,(b - cY) = C - \tfrac{1}{2}K\,(b - cY), \qquad C = d_+ + d_-Y, \tag{4.7}$$

with $d_+ = 1 + b(4 - M_5)$ and $d_- = 1 - c(4 - M_5)$. In the last step $W = (4 - M_5) - \frac{1}{2}K$ was split into its site-local constant and its hop; the constant parts form C. Note $d_- = -(c(4 - M_5) - 1)$ is *minus* the site-local part of D_-, because Y carries the sign. The Pauli–Villars operator is the same operator at unit mass, $D_{\rm PV} = D|_{m=1}$, $C_{\rm PV} = C|_{m=1}$; $D_\pm$ and K do not depend on m.

B. Γ_5-transformation and the exact adjoint

For the Wilson operator $\gamma_5 W \gamma_5 = W^\dagger$, so a solve with $W^\dagger$ is a solve with W between two multiplications by γ_5. The analogue for domain-wall fermions uses

$$\Gamma_5 = \gamma_5 R_5, \qquad (R_5\psi)_s = \psi_{L_s-1-s}. \tag{4.8}$$

Each building block transforms as follows.

1. W, D_+, D_-: they act on every slice alike, so they commute with R_5, and $\gamma_5 W \gamma_5 = W^\dagger$. Hence $\Gamma_5 W \Gamma_5 = W^\dagger$, $\Gamma_5 D_\pm \Gamma_5 = D_\pm^\dagger$.

2. Y: it has no hop in the four dimensions, so $\Gamma_5 Y \Gamma_5$ can be read off Eq. (4.6). γ_5 commutes with $P_\pm$; R_5 turns the hop $s \to s+1$ into $s \to s-1$. So $\Gamma_5 Y \Gamma_5$ has $-P_-$ on the hop down and $-P_+$ on the hop up, and the corner $-m$ moves to the other corner. This is the transpose of Eq. (4.6) in s with $P_\pm$ in place, which is $Y^\dagger$ since $P_\pm^\dagger = P_\pm$ and m is real: $\Gamma_5 Y \Gamma_5 = Y^\dagger$.

3. W and Y do not commute: γ_μ in K anticommutes with γ_5, so $\gamma_\mu P_\pm = P_\mp \gamma_\mu$, and the hop of K exchanges the two projectors of Y.

From $D = D_+ - D_-Y$ it follows that

$$\Gamma_5 D \Gamma_5 = D_+^\dagger - D_-^\dagger Y^\dagger, \qquad \text{whereas} \qquad D^\dagger = D_+^\dagger - Y^\dagger D_-^\dagger, \tag{4.9}$$

the adjoint reversing the order of the product. The difference is a commutator,

$$\Gamma_5 D \Gamma_5 - D^\dagger = Y^\dagger D_-^\dagger - D_-^\dagger Y^\dagger = c\,[Y^\dagger, W^\dagger] = \tfrac{c}{2}\big[K^\dagger, Y^\dagger\big], \tag{4.10}$$

where $D_-^\dagger = cW^\dagger - 1$ and $W^\dagger = (4 - M_5) - \frac{1}{2}K^\dagger$ were used. It vanishes for Shamir's operator ($c = 0$) and is of order c otherwise. Thus $\Gamma_5 D \Gamma_5 \neq D^\dagger$ for Möbius fermions: a multigrid built for D cannot be used for $D^\dagger$ by conjugating with Γ_5. On a 12^4 configuration with stout links ($M_5 = 1$, $m = 0.001$), $\|\Gamma_5 D \Gamma_5 b - D^\dagger b\| / \|D^\dagger b\| = 0.15$ for $(b, c) = (1.5, 0.5)$, 0.22 for $(2, 1)$, and 0 for $(1, 0)$, for a random b.

The exact relation. D_+ and D_- are functions of the same W and commute. Where D_- is invertible, factor it out to the left:

$$D = D_-\,K_D, \qquad K_D = D_-^{-1} D_+ - Y. \tag{4.11}$$

K_D is Γ_5-Hermitian: by item 1, $\Gamma_5 D_-^{-1} D_+ \Gamma_5 = (D_-^\dagger)^{-1} D_+^\dagger$, and this equals $(D_-^{-1} D_+)^\dagger = D_+^\dagger (D_-^\dagger)^{-1}$ because $D_+^\dagger$ and $D_-^\dagger$ commute; with item 2, $\Gamma_5 K_D \Gamma_5 = K_D^\dagger$. Then

$$D^\dagger = K_D^\dagger D_-^\dagger = \Gamma_5 K_D \Gamma_5\, D_-^\dagger = \Gamma_5 D_-^{-1} D\, \Gamma_5 D_-^\dagger, \tag{4.12}$$

and multiplying by $D_-\Gamma_5$ from the left,

$$\boxed{D_-\,\Gamma_5\,D^\dagger = D\,\Gamma_5\,D_-^\dagger} \tag{4.13}$$

This form has no inverse, holds for any m (so also for $D_{\rm PV}$, with the same D_-), and can be checked by applying both sides to a vector. On the configuration above both sides agree to 3×10^{-16} for $(b,c)=(1.5,0.5)$ and $(2,1)$. The Hermitian operator of the Möbius case is $\Gamma_5 K_D=\Gamma_5 D_-^{-1}D$; Ref. [8], Eq. (2.22), defines its Γ_5 as $\gamma_5 R_5 D_-^{-1}$ for that reason. It contains D_-^{-1}, whose cost is discussed in Sec. XI.

C. The site-diagonal part and its inverse

$C=d_++d_-Y$ of Eq. (4.7) is what remains of D when the four-dimensional hops are dropped. It acts within one four-dimensional site, as an $L_s\times L_s$ matrix in s times the identity in color, and it commutes with $P_\pm$ (only Y is involved, and Y maps $P_+\psi$ to $P_+\psi$ and $P_-\psi$ to $P_-\psi$). So it splits into a block on $\psi^+=P_+\psi$ and one on $\psi^-=P_-\psi$. From Eq. (4.6), Y moves ψ^+ from $s-1$ to s and ψ^- from $s+1$ to s, with the factor -1, and the wrap-around gives $+m$:

$$(C\psi)^+_s=\begin{cases} d_+\psi^+_0+m\,d_-\psi^+_{L_s-1}, & s=0,\\ d_+\psi^+_s-d_-\psi^+_{s-1}, & s\ge1,\end{cases} \tag{4.14}$$

a lower bidiagonal matrix with one corner element; for ψ^- the same with $s\to L_s-1-s$ (upper bidiagonal). For $L_s=4$ the block on ψ^+ is

$$C^+=\begin{pmatrix} d_+ & 0 & 0 & m\,d_-\\ -d_- & d_+ & 0 & 0\\ 0 & -d_- & d_+ & 0\\ 0 & 0 & -d_- & d_+\end{pmatrix}. \tag{4.15}$$

The inverse. To solve $C^+x=r$, write the rows $s\ge1$ as $x_s=(r_s+d_-x_{s-1})/d_+$. Starting from an unknown x_0, each x_s is affine in x_0: with $q=d_-/d_+$,

$$x_s=p_s+q^s x_0,\qquad p_0=0,\quad p_s=(r_s+d_-p_{s-1})/d_+, \tag{4.16}$$

which is checked by induction: $(r_s+d_-(p_{s-1}+q^{s-1}x_0))/d_+=p_s+q^sx_0$. The row $s=0$, $d_+x_0+m\,d_-x_{L_s-1}=r_0$, with $x_{L_s-1}=p_{L_s-1}+q^{L_s-1}x_0$, fixes x_0:

$$x_0=\frac{r_0-m\,d_-\,p_{L_s-1}}{d_++m\,d_-\,q^{L_s-1}},\qquad x_s=p_s+q^sx_0. \tag{4.17}$$

For C^- the substitution runs from $s=L_s-1$ downwards. $C^\dagger$ has the transposed blocks: on ψ^+ an upper bidiagonal matrix, solved like C^-, and on ψ^- a lower one. C^{-1} is thus exact, costs $O(L_s)$ per site, and does not depend on the gauge field [24]. It exists unless $d_++m\,d_-q^{L_s-1}=0$.

D. The preconditioned operator

The spectrum of D surrounds the origin (Sec. III), so Krylov methods and polynomial smoothers cannot work on it directly. Ref. [24] multiplies $D\psi=\eta$ from the left by $(D_{\rm PV}C_{\rm PV}^{-1})^\dagger$ and substitutes $\psi=C^{-1}y$:

$$A'y=(D_{\rm PV}C_{\rm PV}^{-1})^\dagger\,\eta,\qquad A'=(D_{\rm PV}C_{\rm PV}^{-1})^\dagger\,D\,C^{-1},\qquad \psi=C^{-1}y. \tag{4.18}$$

Why these two factors:

- DC^{-1} is D with its site-local part divided out; its eigenvalues are much more compact than those of D (Fig. 2: real parts $[-0.22,2.22]$ against $[-1.29,13.2]$), but still on both sides of the origin.

- For $m = 1$, $DC^{-1} = D_{\rm PV}C_{\rm PV}^{-1}$ and $A' = (D_{\rm PV}C_{\rm PV}^{-1})^\dagger(D_{\rm PV}C_{\rm PV}^{-1})$, which is Hermitian and positive definite. For $m < 1$ the two factors differ only in the corners of Eq. (4.5), and A' is a perturbation of a positive operator.
- $D_{\rm PV}^{-1}D$ approximates the overlap operator [21]; $D_{\rm PV}^\dagger$ in place of $D_{\rm PV}^{-1}$ keeps the cost of one application of A' at two applications of D plus the site-local C^{-1} and $C_{\rm PV}^{-1}$.

The result is an operator whose eigenvalues all have positive real part (Fig. 3), with the smallest condition number among those of Ref. [24].

E. Link smearing

The fermion operator may be built from smeared links in place of $U_\mu(x)$; smearing integrates the surrounding gauge field into each link and suppresses the short-distance fluctuations that produce near-zero modes of W. We use the stout projection combined with the APE staples [9]:

$$C_\mu(x) = \sum_{\nu\neq\mu} \rho\Big[U_\nu(x)U_\mu(x+\hat\nu)U_\nu^\dagger(x+\hat\mu) + U_\nu^\dagger(x-\hat\nu)U_\mu(x-\hat\nu)U_\nu(x-\hat\nu+\hat\mu)\Big], \tag{4.19}$$

$$U'_\mu(x) = \exp\Big(\big[C_\mu(x)U_\mu^\dagger(x)\big]_{\rm AT}\Big)\,U_\mu(x), \tag{4.20}$$

where $[\cdots]_{\rm AT}$ is the anti-Hermitian traceless part and ρ a tunable parameter. The step (4.20) is applied repeatedly; "stout $\rho\times n$" means n steps with the same ρ. Only W (through K) sees the smeared links; Y, C and Q do not depend on the gauge field. In HMC the force of the fermion action is taken with respect to the smeared links and carried back to U through the derivative of Eq. (4.20).

F. Neff's α

Neff's form of the domain-wall matrix [25, 26] is $D^{(\alpha)} = DQ$ with a site-local Q, diagonal in the fifth dimension,

$$Q_0 = P_- + \alpha P_+, \qquad Q_s = \alpha \;\; (0 < s < L_s - 1), \qquad Q_{L_s-1} = P_+ + \alpha P_-. \tag{4.21}$$

Properties used below, each read off Eq. (4.21):

- Q is Hermitian, and Q^{-1} is Eq. (4.21) with $\alpha \to 1/\alpha$.
- Q leaves the components $P_-\psi_0$ and $P_+\psi_{L_s-1}$ unchanged, which make up the quark field q; so $D^{(\alpha)}$ and D have the same four-dimensional propagator.
- The solution of $D\psi = \eta$ is $\psi = Q\psi^{(\alpha)}$ with $D^{(\alpha)}\psi^{(\alpha)} = \eta$, since $D^{(\alpha)}\psi^{(\alpha)} = DQ\psi^{(\alpha)} = D\psi$.
- Q is site-local, so the site-diagonal part of DQ is CQ, with $(CQ)^{-1} = Q^{-1}C^{-1}$ and $(CQ)^\dagger = QC^\dagger$.

The factor may be applied to the quark operator alone or to the Pauli–Villars operator as well; we write $D_{\rm PV}^{(\alpha)} = D_{\rm PV}Q$ in the latter case. For $\alpha < 1$ it improves the conditioning of the even-odd operator [25, 26].

V. Even-odd forms

With $D = \begin{pmatrix} C_e & D_{eo} \\ D_{oe} & C_o \end{pmatrix}$, $D_{pq} = -\frac{1}{2}K_{pq}(b - cY)$, the Schur complement on the even sites is $S = C_e - D_{eo}C_o^{-1}D_{oe}$, and $S_{\rm PV}$ is that of $D_{\rm PV}$; note $S^{-1} = \pi D^{-1}\iota$ and $(S_{\rm PV}^\dagger)^{-1} = \pi(D_{\rm PV}^\dagger)^{-1}\iota$. In the same block notation the injection and the projection of the Conventions are

$$\iota y = \begin{pmatrix} y \\ 0 \end{pmatrix}, \qquad \pi\begin{pmatrix} \psi_e \\ \psi_o \end{pmatrix} = \psi_e, \qquad \iota\pi\begin{pmatrix} \psi_e \\ \psi_o \end{pmatrix} = \begin{pmatrix} \psi_e \\ 0 \end{pmatrix}, \tag{5.1}$$

TABLE 1. Multigrid methods for domain-wall fermions: the system solved, the operator that is coarsened, and its coarse stencil, for blocks that contain the whole fifth dimension and are at least two sites wide in each direction (4^4 for HDCG). S is the Schur complement on the even sites.

method	system solved	operator coarsened	coarse stencil
HDCG [18, 16]	even sites, $S^\dagger S$	$S^\dagger S$ (four hops)	81 points
HDCR [19]	whole lattice, $\gamma_5 R_5 D$	$\gamma_5 R_5 D$, $c = 0$ (one hop)	9 points
Boyle and Oare [22]	whole lattice, $D_{\mathrm{PV}}^\dagger D$	$D_{\mathrm{PV}}^\dagger D$ (two hops)	33 points
this note	even sites, $\tilde{A}_R$ or $\tilde{A}_L$	D and D_{PV} apart (one hop each)	9 points each

so $\iota\pi$ sets the odd sites to zero and leaves a field that already vanishes there unchanged. $S^{-1} = \pi D^{-1} \iota$ follows: the solution of $D\psi = \iota\eta_e = (\eta_e, 0)^{\mathsf{T}}$ has $\psi_o = -C_o^{-1} D_{oe} \psi_e$ from the odd rows, and the even rows then read $S\psi_e = \eta_e$. The site-diagonal factor can stand on either side of S, which gives the right and the left form,

$$\tilde{A}_R = (S_{\mathrm{PV}} C_{\mathrm{PV},e}^{-1})^\dagger \, (S\, C_e^{-1}), \qquad \tilde{A}_L = (C_{\mathrm{PV},e}^{-1} S_{\mathrm{PV}})^\dagger \, (C_e^{-1} S). \tag{5.2}$$

$\tilde{A}_R$ is the even-odd counterpart of A'; $C_e^{-1} S$ is the usual left preconditioned even-odd operator, which Boyle and Yamaguchi found inferior to S itself for CG on the normal equation [20]. With α, DQ has Schur complement SQ_e and diagonal block $C_e Q_e$, so $\tilde{A}_R$ is independent of α, whereas

$$\tilde{A}_L^{(\alpha)} = \begin{cases} Q_e \, (C_{\mathrm{PV},e}^{-1} S_{\mathrm{PV}})^\dagger \, Q_e^{-2} \, (C_e^{-1} S) \, Q_e, & \alpha \text{ in } D \text{ and } D_{\mathrm{PV}}, \\ (C_{\mathrm{PV},e}^{-1} S_{\mathrm{PV}})^\dagger \, Q_e^{-1} \, (C_e^{-1} S) \, Q_e, & \alpha \text{ in } D \text{ only}. \end{cases} \tag{5.3}$$

For $D\psi = \eta$, with $\eta'_e = \eta_e - D_{eo} C_o^{-1} \eta_o$, the right-hand side is $\rho_R = (S_{\mathrm{PV}} C_{\mathrm{PV},e}^{-1})^\dagger \eta'_e$ or $\rho_L = (C_{\mathrm{PV},e}^{-1} S_{\mathrm{PV}})^\dagger C_e^{-1} \eta'_e$ (operators with Q when $\alpha \neq 1$); the solution is $\psi_e^{(\alpha)} = C_e^{-1} y$ (right) or y (left), $\psi_o^{(\alpha)} = C_o^{-1}(\eta_o - D_{oe} \psi_e^{(\alpha)})$, $\psi = Q\psi^{(\alpha)}$.

Why both forms. The two forms serve different purposes. The right form is the even-odd counterpart of A', and the factorized correction of Sec. VI is exact for it on the lifted basis (Proposition 1); but Q cancels in it, so it cannot use Neff's α. The left form keeps α and is the one the outer solver uses in practice: the basis is not exact for it, yet it converges within a few iterations of the right form, and with α it can save iterations. The measurements of Sec. XII are on the left form unless stated.

VI. The factorized coarse-grid correction and the prolongation

A. Why factorize

Solving on the even sites halves the problem and improves the conditioning. The Schur complement is a two-hop operator, so the methods in the literature make different choices: HDCG coarsens the squared Schur complement, with a larger coarse stencil [18], and the methods with a nearest-neighbor or two-hop coarse operator coarsen an operator on the whole lattice: $\gamma_5 R_5 D$ (HDCR [19]) or $D_{\mathrm{PV}}^\dagger D$ [21, 22]. The factorized correction combines the two: the outer operator is the even-odd one, while the coarse operators are those of D and D_{PV} themselves (Table 1).

The point counts follow from the hops: an operator that reaches k hops on the fine lattice couples coarse sites whose coordinates differ by at most 1 in each direction and in at most k directions, which for $k = 1, 2$ gives $1 + 8 = 9$ and $1 + 8 + 24 = 33$ points and for $k \geq 4$ all $3^4 = 81$. By $S^{-1} = \pi D^{-1} \iota$, the inverse of the right form is

$$\tilde{A}_R^{-1} = C_e \, S^{-1} \, (S_{\mathrm{PV}}^\dagger)^{-1} \, C_{\mathrm{PV},e}^\dagger = C_e \, \pi D^{-1} \iota \; \pi (D_{\mathrm{PV}}^\dagger)^{-1} \iota \; C_{\mathrm{PV},e}^\dagger, \tag{6.1}$$

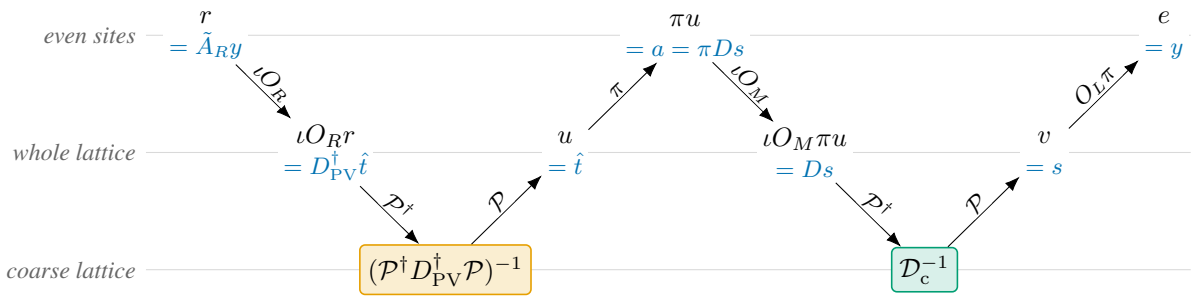


FIG. 4. One factorized coarse correction, Eq. (6.2) read from the right, on its three lattices (the Pauli–Villars coarse solve in orange, the quark one in green): $u = \mathcal{P}(\mathcal{P}^\dagger D_{\mathrm{PV}}^\dagger \mathcal{P})^{-1}\mathcal{P}^\dagger \iota O_R r$ is the result of the first coarse solve, that of the adjoint Pauli–Villars operator, and its even part, with O_M, is the right-hand side of the second, that of the quark operator, whose result is $v = \mathcal{P}\mathcal{D}_{\mathrm{c}}^{-1}\mathcal{P}^\dagger \iota O_M \pi u$; $e = O_L \pi v$. In blue, after "=", the values for $r = \tilde{A}_R y$ in the right form with exact coarse solves and $s, \hat{t} \in \operatorname{range} \mathcal{P}$ (Proposition 1; $O_R = C_{\mathrm{PV},e}^\dagger$, $O_M = 1$, $O_L = C_e$): the first solve returns the lift $\hat{t}$, whose even part is $a = \pi D s$, the second returns the lift s, and the correction returns y.

TABLE 2. Site-local factors of the correction (6.2). They follow from $\tilde{A}^{-1}$ with $S^{-1} = \pi D^{-1} \iota$; with α the site-diagonal blocks are $C_e Q_e$ and $Q_e C_{\mathrm{PV},e}^\dagger$.

form	O_R	O_M	O_L
right, any α	$C_{\mathrm{PV},e}^\dagger$	1	C_e
left, α in D only	1	$C_e Q_e C_{\mathrm{PV},e}^\dagger$	Q_e^{-1}
left, α in D and D_{PV}	Q_e^{-1}	$C_e Q_e^2 C_{\mathrm{PV},e}^\dagger$	Q_e^{-1}

a product of two inverses of one-hop operators with site-local factors. The Galerkin operator of $\tilde{A}_R$ itself, a four-hop operator on the fine lattice, would have a much larger coarse stencil; the Galerkin operators of D and D_{PV} are nearest-neighbor, because D is nearest-neighbor in four dimensions and every aggregate contains the whole fifth dimension. We therefore replace each inverse by its Galerkin approximation $\mathcal{P}(\mathcal{P}^\dagger X \mathcal{P})^{-1}\mathcal{P}^\dagger$. For both forms and any α, the correction of a residual r on the even sites is

$$e = O_L\, \pi\mathcal{P}\, \mathcal{D}_{\mathrm{c}}^{-1} \mathcal{P}^\dagger \iota \;\; O_M \;\; \pi\mathcal{P}\, \left(\mathcal{P}^\dagger D_{\mathrm{PV}}^\dagger \mathcal{P}\right)^{-1} \mathcal{P}^\dagger \iota \;\; O_R\, r, \qquad \mathcal{D}_{\mathrm{c}} = \mathcal{P}^\dagger D \mathcal{P}, \tag{6.2}$$

with the site-local factors of Table 2. The coarse operators are always those of the plain operators D and D_{PV} (no α), so one setup serves both forms and every α. Figure 4 follows a residual through Eq. (6.2): it moves between the even sites, the whole lattice and the coarse lattice twice, and the result of the first coarse solve, taken on the even sites, is the right-hand side of the second.

B. Aggregation and prolongation

The fine lattice is divided into rectangular four-dimensional blocks; each block contains the whole fifth dimension and becomes one coarse site. Each test vector v is split into its Γ_5 eigencomponents $v_\pm = \frac{1}{2}(1 \pm \Gamma_5)v$, and within each block the $+$ parts and the $-$ parts are orthonormalized separately. With n_v test vectors a coarse site carries $2n_v$ components, the first n_v of positive and the last n_v of negative Γ_5-chirality. Writing $\Sigma = \operatorname{diag}(\mathbf{1}_{n_v}, -\mathbf{1}_{n_v})$ on each coarse site,

$$\Gamma_5 \mathcal{P} = \mathcal{P}\Sigma, \tag{6.3}$$

so that $\mathcal{P}^\dagger(\Gamma_5 X \Gamma_5)\mathcal{P} = \Sigma\, \mathcal{P}^\dagger X \mathcal{P}\, \Sigma$ for any fine operator X; in particular $\mathcal{P}^\dagger D_{\mathrm{PV}}^\dagger \mathcal{P} = \Sigma \mathcal{D}_{\mathrm{c}}^{\mathrm{PV}} \Sigma$ with $\mathcal{D}_{\mathrm{c}}^{\mathrm{PV}} = \mathcal{P}^\dagger D_{\mathrm{PV}} \mathcal{P}$ whenever $\Gamma_5 D_{\mathrm{PV}} \Gamma_5 = D_{\mathrm{PV}}^\dagger$, i.e. for $c = 0$. Because $v_\pm(L_s - 1 - s) = \pm\gamma_5 v_\pm(s)$, each part is fixed by half of the fifth dimension, so $\mathcal{P}$ need only be stored for $s < L_s/2$. On the even sites the transfer operators are $\mathcal{P}^\dagger \iota$ and $\pi\mathcal{P}$.

C. The basis $\{s, \hat{t}\}$

The prolongation is built from vectors y_j on the even sites that approximate low modes of $\tilde{A}_R$ (Sec. VIII). Since $\mathcal{P}$ and the coarse operators act on the whole lattice, each y is extended, or lifted, to the whole lattice, the odd sites being filled so that the fine operator maps the lifted vector to the even-odd quantity, as ψ_o is reconstructed from ψ_e after an even-odd solve; and since the correction (6.2) contains two Galerkin solves, with $D^\dagger_{\rm PV}$ and with D, each exact only on vectors in $\mathrm{range}\,\mathcal{P}$, each y is lifted in two ways, one for each solve:

$$s = \big(C_e^{-1}y,\ -C_o^{-1}D_{oe}C_e^{-1}y\big), \qquad Ds = \iota\, SC_e^{-1}y, \tag{6.4}$$

$$\hat{t} = \big(a,\ -(C^\dagger_{{\rm PV},o})^{-1}(D_{{\rm PV},eo})^\dagger a\big), \quad a = \pi Ds, \qquad D^\dagger_{\rm PV}\hat{t} = \iota\, S^\dagger_{\rm PV}a, \tag{6.5}$$

where the right-hand columns follow from the block form of D and of $D^\dagger_{\rm PV}$, whose odd-even block is $(D_{{\rm PV},eo})^\dagger$ and whose odd diagonal block is $C^\dagger_{{\rm PV},o}$. The images of s under D and of $\hat{t}$ under $D^\dagger_{\rm PV}$ thus vanish on the odd sites. The first coarse solve of the correction returns $\hat{t}$ and the second s (Proposition 1); $\hat{t}$ is made from Ds, so the two lifts follow the order of the solves. With the y alone in the basis, each factor of the correction would be an approximation and the errors of the two would multiply. The structure is familiar from the block preconditioners of the incompressible Navier–Stokes equations [33]: the velocity is eliminated, as the odd sites are here, and the pressure Schur complement is approximated by a product of simpler operators, each solved by its own multigrid; in the least-squares commutator preconditioner, $(BB^\mathsf{T})^{-1}(BFB^\mathsf{T})(BB^\mathsf{T})^{-1}$, two Poisson-like inverses. The product is only as good as each factor on the vectors it is applied to, which here are $\hat{t}$ and s. The $2n$ vectors $\{s_j, \hat{t}_j\}$ are the test vectors, $n_v = 2n$.

Proposition 1 (Exactness, right form)**.** *Let $s, \hat{t} \in \mathrm{range}\,\mathcal{P}$, let $\mathcal{D}_{\rm c}$ and $\mathcal{P}^\dagger D^\dagger_{\rm PV}\mathcal{P}$ be invertible, and let both coarse systems in Eq.* (6.2) *be solved exactly with the exact adjoint $\mathcal{P}^\dagger D^\dagger_{\rm PV}\mathcal{P} = (\mathcal{D}_{\rm c}^{\rm PV})^\dagger$. Then the right-form correction of $r = \tilde{A}_R y$ is $e = y$.*

Proof. For any invertible $\mathcal{P}^\dagger X\mathcal{P}$, $\mathcal{P}(\mathcal{P}^\dagger X\mathcal{P})^{-1}\mathcal{P}^\dagger X$ is the identity on $\mathrm{range}\,\mathcal{P}$. Now $O_R r = C^\dagger_{{\rm PV},e}\tilde{A}_R y = S^\dagger_{\rm PV}SC_e^{-1}y = S^\dagger_{\rm PV}a$, and $\iota S^\dagger_{\rm PV}a = D^\dagger_{\rm PV}\hat{t}$ by Eq. (6.5). The first coarse solve returns $\pi\hat{t} = a = \pi Ds$. With $O_M = 1$ the second has $\mathcal{P}^\dagger\iota\pi Ds = \mathcal{P}^\dagger Ds$ on the right, since Ds vanishes on the odd sites, and returns $\pi s = C_e^{-1}y$; $O_L = C_e$ gives y. □

Remark 1 (Conditions in practice). The proposition needs (i) $s, \hat{t}$ in the range of the stored $\mathcal{P}$, which holds only up to the rounding of the reduced-precision storage of $\mathcal{P}$; (ii) the exact adjoint of the Pauli–Villars coarse factor—the Σ-adjoint $\Sigma\mathcal{D}_{\rm c}^{\rm PV}\Sigma$ of Sec. VII equals it only if $\Gamma_5 D_{\rm PV}\Gamma_5 = D^\dagger_{\rm PV}$, which by Eq. (4.10) fails for $c \neq 0$; (iii) exact coarse solves, whereas they are stopped at a loose tolerance. Numerically, with both coarse factors solved on their exact adjoints to $|r|^2/|b|^2 < 10^{-24}$ in double precision and $\mathcal{P}$ in half precision as the solver stores it, the right-form correction of $r = \tilde{A}_R y_j$ returns y_j to $\|e - y_j\|/\|y_j\| = 3\times10^{-4}$–$5\times10^{-4}$ on $4^3\times8$ (2^4 aggregates, $M_5 = 1.8$, $m = 0.05$) and 3×10^{-4}–2.5×10^{-3} on 12^4 (setup S, $m = 0.001$), against 3–37 for random vectors: the proposition holds to the rounding of the stored $\mathcal{P}$, so (i) costs about 10^{-3}. (ii) costs nothing measurable (Sec. XII.G), and (iii) helps: with GCR(2), accurate coarse solves need more iterations or stagnate (Secs. XII.E, XII.H).

Remark 2 (Left form). For the left form with α in both operators, the same argument requires the quark lift s' of $Q_e y'$ and the Pauli–Villars lift $\hat{t}'$ of $u = C^{\dagger\,-1}_{{\rm PV},e}Q_e^{-2}C_e^{-1}SQ_e y'$ to lie in $\mathrm{range}\,\mathcal{P}$ (Table 2 gives $O_M u = SQ_e y'$). For $\alpha = 1$ and $y' = C_e^{-1}y$ one has $s' = s$, but $u = C^{\dagger\,-1}_{{\rm PV},e}C_e^{-1}a$ differs from a by a site-local matrix that mixes s and spin, so $\hat{t}'$ is in general not in the span of the right-form vectors, and the correction is not exact for the left form with the basis above. Numerically (as in the previous remark, $\alpha = 1$), for $r = \tilde{A}_L y'$ with $y' = C_e^{-1}y_j$ the correction misses y' by $\|e - y'\|/\|y'\| = 0.05$–0.18 on $4^3\times8$ and 1.6–24 on 12^4, the latter not much better than random vectors (8–35). The left form converges nonetheless: it has the spectrum of the right form and needs one to five iterations more or fewer

(Secs. XII.C, XII.E, XII.H), so the exactness of Proposition 1 is not needed for the cycle to work; whether a basis exact for both forms at once exists is left open.

D. The coarse operators

$\mathcal{D}_\mathrm{c} = \mathcal{P}^\dagger D \mathcal{P}$ and $\mathcal{D}_\mathrm{c}^\mathrm{PV} = \mathcal{P}^\dagger D_\mathrm{PV} \mathcal{P}$ are assembled from the hops of the fine operator between neighboring blocks and from its block-internal part: on each coarse site a $2n_v \times 2n_v$ diagonal block and one link block per direction, nine blocks in all. The adjoint factor $\mathcal{P}^\dagger D_\mathrm{PV}^\dagger \mathcal{P}$ is not assembled; it is applied either exactly, as the conjugate transpose $(\mathcal{D}_\mathrm{c}^\mathrm{PV})^\dagger$ of the stored blocks, or approximately as $\Sigma \mathcal{D}_\mathrm{c}^\mathrm{PV} \Sigma$. When all extents of the coarse lattice are even, the coarse hops connect coarse sites of opposite parity and both coarse operators admit an even-odd decomposition of their own.

VII. The coarse solvers

Both coarse systems are solved approximately, to a loose relative residual or a maximum number of iterations, in reduced precision; reaching the tolerance is not required. They are solved in sequence: first the Pauli–Villars factor, whose solution enters the right-hand side of the quark factor. This section lists the options; the default, used for all results unless stated, is BiCGStab on the even-odd form of the quark factor and MINRES on the Σ-Hermitian form of the Pauli–Villars factor, both to $|r|^2/|b|^2 < 10^{-2}$.

A. The quark factor

Write $\mathcal{D}_\mathrm{c} = \begin{pmatrix} A_{ee} & B_{eo} \\ B_{oe} & A_{oo} \end{pmatrix}$ on the coarse lattice. The system $\mathcal{D}_\mathrm{c} c = g$ is solved through its Schur complement,

$$\left(A_{ee} - B_{eo} A_{oo}^{-1} B_{oe}\right) c_e = g_e - B_{eo} A_{oo}^{-1} g_o, \qquad c_o = A_{oo}^{-1}(g_o - B_{oe} c_e), \tag{7.1}$$

with BiCGStab or truncated flexible GCR, preconditioned by the exact inverses of the diagonal blocks A_{ee}^{-1}. $\mathcal{D}_\mathrm{c}$ is not Hermitian and $\Sigma\mathcal{D}_\mathrm{c}$ is Hermitian only for $c = 0$, so a non-Hermitian Krylov method is used.

B. The Pauli–Villars factor

The system is $(\mathcal{P}^\dagger D_\mathrm{PV}^\dagger \mathcal{P})\, w = g$. Two treatments are available.

Exact adjoint. Solve $(\mathcal{D}_\mathrm{c}^\mathrm{PV})^\dagger w = g$ with a non-Hermitian Krylov method, preconditioned by the inverses of the adjoint diagonal blocks, $(A_{xx}^\dagger)^{-1} = (A_{xx}^{-1})^\dagger$.

Σ-Hermitian form. Replace the adjoint by $\Sigma\mathcal{D}_\mathrm{c}^\mathrm{PV}\Sigma$ and solve the equivalent system

$$\mathcal{H}_\mathrm{c}\, w = \Sigma g, \qquad \mathcal{H}_\mathrm{c} = \mathcal{D}_\mathrm{c}^\mathrm{PV}\Sigma, \tag{7.2}$$

with MINRES. $\mathcal{H}_\mathrm{c}$ is Hermitian (indefinite) exactly when $(\mathcal{D}_\mathrm{c}^\mathrm{PV})^\dagger = \Sigma\mathcal{D}_\mathrm{c}^\mathrm{PV}\Sigma$, i.e. for $c = 0$ by Eqs. (4.10) and (6.3); for $c \neq 0$ MINRES is applied to a non-Hermitian matrix and has no guarantee. On the coarse space the non-Hermitian part is a few percent, up to 14% in the directions of the smallest singular values, and linear in c; MINRES on $\mathcal{H}_\mathrm{c}$ nevertheless needs the same outer iterations at $c = 0.5$ as for the Shamir operator and is better than the exact adjoint with BiCGStab (Sec. XII.G). The absolute-value block preconditioner and the additive level preconditioner below, also exact only for $c = 0$, are used as they are; their losses were not measured separately.The preconditioner must be Hermitian positive definite; we use the absolute value of the diagonal blocks [27], $|\mathcal{H}_{xx}|^{-1} = V|\Lambda|^{-1}V^\dagger$ with $\frac{1}{2}(\mathcal{H}_{xx} + \mathcal{H}_{xx}^\dagger) = V\Lambda V^\dagger$ on each coarse site. On a coarse lattice with even extents, the Schur complement of $\mathcal{H}_\mathrm{c}$ on the even sites is $S_\mathrm{c}\Sigma_e$, with S_c the Schur complement of $\mathcal{D}_\mathrm{c}^\mathrm{PV}$; one solves for $u_e = \Sigma w_e$ on the even sites, completes u on the odd sites as in Eq. (7.1), and sets $w = \Sigma u$.

C. Levels below the coarse lattice

Both factors may be preconditioned by further levels. The coarse sites are grouped into super-blocks; on each super-block, local vectors are formed from the coarse test vectors $\mathcal{P}^\dagger s_j$ and $\mathcal{P}^\dagger \hat{t}_j$, split by Σ and orthonormalized (Gram–Schmidt applied twice); the level operators are obtained by probing the nearest-neighbor stencil; the last level is inverted densely. With Q_ℓ the prolongation into level $\ell - 1$ and B_ℓ the diagonal blocks of level ℓ:

Quark factor. A V-cycle with the coarse correction followed by one block-Jacobi sweep on the updated residual, without pre-smoothing,

$$z = Q_{\ell+1} M_{\ell+1} Q_{\ell+1}^\dagger r, \qquad M_\ell r = z + B_\ell^{-1}(r - \mathcal{D}_\ell z); \tag{7.3}$$

on the coarse lattice itself the same multiplicative combination is used, and with the even-odd coarse form its even part preconditions the Schur system. The level test vectors can optionally be refined by a few GCR steps preconditioned by the levels already built.

Pauli–Villars factor (Σ-Hermitian form only). The additive Hermitian positive preconditioner

$$M_\ell = |B_\ell|^{-1} + Q_{\ell+1} M_{\ell+1} Q_{\ell+1}^\dagger, \qquad M_{\ell_{\max}} = |\mathcal{H}_{\ell_{\max}}|^{-1}, \tag{7.4}$$

in the spirit of absolute-value preconditioning [27].

VIII. Setup

The setup produces the vectors $y_1, \ldots, y_n$ on the even sites, then the basis $\{s_j, \hat{t}_j\}$, $\mathcal{P}$, and the coarse operators. It always uses the right form; the same setup serves the left forms and every α (Sec. VI). It has two stages: a Chebyshev filter finds approximate low modes without any multigrid, and a bootstrap then improves them with the cycle built from them, so that they also cover what the first coarse space misses.

A. Chebyshev-filtered subspace iteration

The vectors approximate the right singular vectors of the smallest singular values of $\tilde{A}_R$, i.e. eigenvectors of the Hermitian positive operator $H_n = \tilde{A}_R^\dagger \tilde{A}_R$. With the Chebyshev polynomial T_d and $\ell(x) = \big(2x - (\lambda_{\rm hi} + \lambda_{\rm lo})\big)/(\lambda_{\rm hi} - \lambda_{\rm lo})$, $|T_d(\ell(x))| \le 1$ on $[\lambda_{\rm lo}, \lambda_{\rm hi}]$ while components below $\lambda_{\rm lo}$ grow; $T_d(\ell(H_n))$ is applied by the three-term recurrence.

Algorithm 1. LOWMODES: Chebyshev-filtered subspace iteration on $H_n = \tilde{A}_R^\dagger \tilde{A}_R$.

1: $X \leftarrow n$ random vectors on the even sites
2: $\lambda_{\rm hi} \leftarrow$ a bound slightly above $\lambda_{\max}(H_n)$ (power iteration); $\lambda_{\rm lo} \leftarrow$ a small fraction of $\lambda_{\rm hi}$
3: **for** each round **do**
4: $d \leftarrow$ a degree proportional to $\sqrt{\lambda_{\rm hi}/\lambda_{\rm lo}}$, within fixed bounds
5: $X \leftarrow T_d(\ell(H_n))\, X$; orthonormalize X through its Gram matrix
6: $G \leftarrow X^\dagger H_n X = V \Theta V^\dagger$; $X \leftarrow XV$ ▷ Rayleigh–Ritz
7: $\lambda_{\rm lo} \leftarrow \theta_{\max}$ ▷ largest Ritz value
8: **end for**
9: **return** X, Θ

B. Bootstrap

Each bootstrap round builds the basis and the two-grid cycle of Sec. IX from the current y, replaces every y_j by the result of a fixed small number of FGMRES steps on $\tilde{A}_R x = y_j$ from $x = 0$ preconditioned by

that cycle, and orthonormalizes the new y through their Gram matrix ($y \leftarrow yV\Lambda^{-1/2}$ with $y^\dagger y = V\Lambda V^\dagger$). The cycle damps the components it already handles, so the new vectors are enriched in what it does not.

Algorithm 2. Setup of the two-grid method.

1: $y \leftarrow \textsc{LowModes}(H_n)$
2: **for** each bootstrap round **do**
3: build $\mathcal{P}$ from $\{s_j, \hat{t}_j\}$ of y and the cycle $\mathcal{M}$ on it
4: **for** $j = 1, \ldots, n$ **do**
5: $y_j \leftarrow$ a few FGMRES steps on $\tilde{A}_R x = y_j$, preconditioned by $\mathcal{M}$
6: **end for**
7: orthonormalize y
8: **end for**
9: compute $s_j, \hat{t}_j$ by Eqs. (6.4)–(6.5); split by Γ_5; orthonormalize per block $\rightarrow \mathcal{P}$
10: assemble $\mathcal{D}_c$, $\mathcal{D}_c^{\mathrm{PV}}$, the inverses of their diagonal blocks, their even-odd forms
11: optionally, build the levels below from $\mathcal{P}^\dagger s_j$, $\mathcal{P}^\dagger \hat{t}_j$

IX. The Chebyshev smoother and the cycle

A. Smoother

The smoother is ν steps of Chebyshev iteration [28, 29] for $\tilde{A}x = b$ from $x = 0$, i.e. a fixed polynomial $x = p(\tilde{A})b$ of degree $\nu - 1$. It is defined by an ellipse with real foci $[\lambda_{\mathrm{lo}}, \lambda_{\mathrm{hi}}]$: λ_{hi} slightly above an estimate of the spectral radius of $\tilde{A}$ from a power iteration, and $\lambda_{\mathrm{lo}} = \lambda_{\mathrm{hi}}/\kappa_s$ for a chosen ratio κ_s. With $\theta = \frac{1}{2}(\lambda_{\mathrm{hi}} + \lambda_{\mathrm{lo}})$, $\delta = \frac{1}{2}(\lambda_{\mathrm{hi}} - \lambda_{\mathrm{lo}})$, $\sigma = \theta/\delta$, $\varrho_0 = 1/\sigma$, $r_0 = b$ and $d_0 = r_0/\theta$, step k is

$$x \leftarrow x + d_k, \quad r_{k+1} = r_k - \tilde{A}d_k, \quad \varrho_{k+1} = \frac{1}{2\sigma - \varrho_k}, \quad d_{k+1} = \varrho_{k+1}\varrho_k\, d_k + \frac{2\varrho_{k+1}}{\delta}\, r_{k+1}, \tag{9.1}$$

the last step only updating x; ν steps cost $\nu - 1$ applications of $\tilde{A}$. Only a real focal interval is used. The ratio κ_s is not derived from the operator and has to be chosen for it: the polynomial damps the part of the spectrum above λ_{lo} and leaves the rest to the coarse correction, and too wide or too narrow an interval can make the outer solve stagnate. The two setups of Sec. XII need $\kappa_s \approx 10$ ($M_5 = 1$, stout) and $\kappa_s \approx 30$ ($M_5 = 1.8$); the first stagnates with the second's interval, the second needs nearly three times the iterations with the first's (Sec. XII.H).

B. Cycle

Algorithm 3. Two-grid cycle $y = \mathcal{M}b$ for $\tilde{A}$.

1: $y \leftarrow p(\tilde{A})\, b$ ▷ pre-smoothing
2: $r \leftarrow b - \tilde{A}y$
3: $y \leftarrow y + e(r)$ ▷ correction (6.2)
4: $r \leftarrow b - \tilde{A}y$
5: $y \leftarrow y + p(\tilde{A})\, r$ ▷ post-smoothing

One cycle costs 2ν applications of $\tilde{A}$ (two smoothings and two residuals) plus the two coarse solves. Because the coarse solves are Krylov iterations stopped by a residual test, $\mathcal{M}$ is not a fixed linear map.

X. The outer solver

The outer iteration solves $\tilde{A}y = \rho$ with the cycle as right preconditioner, and ψ is reconstructed as in Sec. V. Since $\mathcal{M}$ varies from one application to the next, the outer method has to be flexible: FGMRES or flexible GCR, which keep the preconditioned directions; BiCGStab with the cycle as preconditioner is also available but has no flexible counterpart of its convergence theory. Flexibility alone gives no convergence guarantee. For a fixed $\mathcal{M}$, restarted or truncated methods converge for every restart length if the field of values of $\tilde{A}\mathcal{M}$ lies in the right half-plane [23]; otherwise only the full method is covered, by the spectrum and the pseudospectra of $\tilde{A}\mathcal{M}$ [15, 31], and a truncated one can stagnate [32]. For the present cycle the field of values is not in the right half-plane, the pseudospectra do bound the full method, and GCR(2) needs the coarse solves inexact: with accurate ones it needs more iterations or stagnates (Secs. XII.B–XII.E). Block variants treat several right-hand sides together. Convergence is monitored on the even-odd system; the residual of the original equation, $\|D\psi - \eta\|/\|\eta\|$, is evaluated with the plain D after reconstruction. The reference solver is CG on the normal equation of the even-odd operator $C_e^{-1}S$ (CGNE), which is robust but squares the condition number; it runs in double precision or in mixed precision, CG in single precision with reliable updates in double precision. For the systems with $\tilde{A}_R^\dagger$ (the normal equations of HMC) RutticeQCD uses the cycle with D and D_{PV} exchanged on the same $\mathcal{P}$ and the same smoother interval; with exact coarse solves it is the adjoint of $\mathcal{M}$ (Sec. XII.B). How well it converges is an open problem (Sec. XIII).

XI. Solving with $D^\dagger$

HMC with the ratio action $\det(D^\dagger D)/\det(D_{\mathrm{PV}}^\dagger D_{\mathrm{PV}})$ needs $x = (D^\dagger D)^{-1}b$ at every force evaluation, that is one solve with $D^\dagger$ and one with D. This section derives the system that the solve with $D^\dagger$ becomes on the even sites, shows why the exact relation of Sec. IV.B, which replaces the shortcut $D^\dagger = \Gamma_5 D \Gamma_5$ of Wilson fermions, cannot be used directly, and asks which prolongation the solve with $D^\dagger$ needs. Everything here is at $\alpha = 1$.

A. The system on the even sites

The block form of $D^\dagger$ is

$$D^\dagger = \begin{pmatrix} C_e^\dagger & D_{oe}^\dagger \\ D_{eo}^\dagger & C_o^\dagger \end{pmatrix}, \tag{11.1}$$

since the (e,o) block of $D^\dagger$ is the adjoint of the (o,e) block of D. Eliminating the odd sites of $D^\dagger w = r$ in the same way as for D (Sec. V) gives

$$\left(C_e^\dagger - D_{oe}^\dagger (C_o^\dagger)^{-1} D_{eo}^\dagger\right) w_e = r_e - D_{oe}^\dagger (C_o^\dagger)^{-1} r_o, \tag{11.2}$$

and the matrix on the left is $S^\dagger$, the adjoint of the Schur complement of D: $S^\dagger = C_e^\dagger - D_{oe}^\dagger (C_o^\dagger)^{-1} D_{eo}^\dagger$ term by term. Write

$$X = S C_e^{-1}, \qquad X_{\mathrm{PV}} = S_{\mathrm{PV}} C_{\mathrm{PV},e}^{-1}, \qquad \text{so} \qquad \tilde{A}_R = X_{\mathrm{PV}}^\dagger X \tag{11.3}$$

by Eq. (5.2). With $S^\dagger = C_e^\dagger X^\dagger$, the even-site equation is $X^\dagger w_e = (C_e^\dagger)^{-1} r_e'$, $r_e' = r_e - D_{oe}^\dagger (C_o^\dagger)^{-1} r_o$. Putting $w_e = X_{\mathrm{PV}} z$,

$$X^\dagger X_{\mathrm{PV}}\, z = (C_e^\dagger)^{-1} r_e', \qquad X^\dagger X_{\mathrm{PV}} = \left(X_{\mathrm{PV}}^\dagger X\right)^\dagger = \tilde{A}_R^\dagger, \tag{11.4}$$

and $w_o = (C_o^\dagger)^{-1}(r_o - D_{eo}^\dagger w_e)$. Thus the solve with $D^\dagger$ is a solve with $\tilde{A}_R^\dagger$, and $\tilde{A}_R^\dagger$ is the right form (5.2) with the roles of D and D_{PV} exchanged: the same code with its two operators swapped. Its inverse, as Eq. (6.1),

$$(\tilde{A}_R^\dagger)^{-1} = X_{\mathrm{PV}}^{-1} (X^\dagger)^{-1} = C_{\mathrm{PV},e}\, S_{\mathrm{PV}}^{-1}\, (S^\dagger)^{-1}\, C_e^\dagger = C_{\mathrm{PV},e}\, \pi D_{\mathrm{PV}}^{-1} \iota\, \pi (D^\dagger)^{-1} \iota\, C_e^\dagger, \tag{11.5}$$

is again a product of two inverses of one-hop operators, now first $D^\dagger$ and then $D_{\rm PV}$, and the factorized correction (6.2) carries over with $O_R = C_e^\dagger$, $O_M = 1$, $O_L = C_{{\rm PV},e}$ and the coarse factors $\mathcal{P}^\dagger D^\dagger \mathcal{P}$ (first) and $\mathcal{P}^\dagger D_{\rm PV} \mathcal{P}$ (second).

B. The exact adjoint and why it cannot be used directly

By Sec. IV.B, $\Gamma_5 D \Gamma_5 \neq D^\dagger$ for $c \neq 0$, and the exact relation Eq. (4.13) gives $D^\dagger = \Gamma_5 D_-^{-1} D\, \Gamma_5 D_-^\dagger$. Two consequences would make one multigrid serve both D and $D^\dagger$:

- *Null vectors.* If $Dv \approx 0$, then $D^\dagger w \approx 0$ for

$$w = (D_-^\dagger)^{-1} \Gamma_5 v, \tag{11.6}$$

  because $D^\dagger w = \Gamma_5 D_-^{-1} D\, \Gamma_5 D_-^\dagger w = \Gamma_5 D_-^{-1} D v$. For $c = 0$, $D_- = -1$ and $w \propto \Gamma_5 v$, the Wilson case.

- *Solves.* $D^\dagger w = r$ is equivalent to $Dz = D_- \Gamma_5 r$, $w = (D_-^\dagger)^{-1} \Gamma_5 z$: a solve with D, by the multigrid of D, followed by a solve with $D_-^\dagger$.

Both need an inverse of D_-.

Why D_- cannot be inverted. $D_- = cW - 1 = c\,(W - 1/c)$, and W is the Wilson operator of mass $-M_5$ (Eq. (4.2)). So D_- is, up to the factor c, the Wilson operator of mass $m_0 = -M_5 - 1/c$: -3 in setup S ($M_5 = 1$, $c = 0.5$) and -3.8 in setup B. Its free eigenvalues (Sec. IV)

$$m_0 + \sum_\mu (1 - \cos p_\mu) \pm i \Big(\sum_\mu \sin^2 p_\mu \Big)^{1/2} \tag{11.7}$$

have real parts filling $[m_0, m_0 + 8]$, here $[-3, 5]$. The eigenvalues with real part near 0 are those with $\sum_\mu (1 - \cos p_\mu) \approx 3$, between the doubler corners with two and with four momenta at π, and they have imaginary parts of both signs: the spectrum surrounds the origin. A Krylov method builds polynomials p with $p(0) = 1$ that should be small on the spectrum; when the spectrum surrounds 0 no such polynomial of low degree exists (Sec. III), and GCR or BiCGStab make no progress. The gauge field does not change this picture. On the A30 (12^4, as above), GCR(8) on D_- and on $D_-^\dagger$ stops at $|r|^2/|b|^2 = 0.74$ (S) and 0.96 (B) after 20000 iterations, and BiCGStab diverges. The inverse of D_- is thus harder than the solve with D it would serve, and neither route above is practical.

What remains is the structure of Eq. (11.6). On smooth fields ($p \approx 0$) the free D_- is the number $-(1 + cM_5)$, so for the smooth part of a low mode v, $w \approx -\Gamma_5 v/(1 + cM_5)$: $\Gamma_5 v$ points along a null vector of $D^\dagger$ up to its rough components.

C. Which prolongation the solve with $D^\dagger$ needs

The correction for $\tilde{A}_R^\dagger$ is exact on the analogue of Proposition 1, with the lifts of Eqs. (6.4) and (6.5) made with D and $D_{\rm PV}$ exchanged: for a vector y' on the even sites,

$$s' = \big(C_{{\rm PV},e}^{-1} y',\ -C_{{\rm PV},o}^{-1} D_{{\rm PV},oe} C_{{\rm PV},e}^{-1} y' \big), \qquad \hat{t}' = \big(a',\ -(C_o^\dagger)^{-1} D_{eo}^\dagger a' \big), \quad a' = \pi D_{\rm PV} s', \tag{11.8}$$

so that $D_{\rm PV} s' = \iota S_{\rm PV} C_{{\rm PV},e}^{-1} y'$ and $D^\dagger \hat{t}' = \iota S^\dagger a'$. If $r = \tilde{A}_R^\dagger y'$ and $s', \hat{t}' \in \operatorname{range} \mathcal{P}$, the correction returns y', by the proof of Proposition 1 with the roles exchanged. The question is which y'.

The test vectors y_j of the setup approximate the right singular vectors of $\tilde{A}_R$ with small singular values: the filter works on $\tilde{A}_R^\dagger \tilde{A}_R$. The right singular vectors of $\tilde{A}_R^\dagger$ are the *left* singular vectors of $\tilde{A}_R$: if $\tilde{A}_R v = \sigma u$ and $\tilde{A}_R^\dagger u = \sigma v$, then u is a right singular vector of $\tilde{A}_R^\dagger$ with the same σ. So the y' needed are the $u_j = \tilde{A}_R y_j/\sigma_j$, which the setup does not provide; computing them as $\tilde{A}_R y_j$ multiplies the error of y_j in the high modes by $\sigma_{\rm high}/\sigma_j$, which is why that basis is worse. (Measured: worse than the y_j

TABLE 3. GCR(8) iterations of the solve with $\tilde{A}_R^\dagger$ for prolongations made from the test vectors y of the setup of $\tilde{A}_R$, without a setup of their own; the first row is the solve with $\tilde{A}_R$ for comparison. "stall": not converged in 1000 ($|r|^2$ at the end in brackets); OOM: out of the 24 GB of device memory of the A30.

	12^4		16^4		24^4	
prolongation	S	B	S	B	S	B
$\tilde{A}_R$ with $\mathcal{P}(y)$	7	9	7	9	9	13
$\mathcal{P}(y)$, shared	stall (10^{-6})	14	10	12	21	20
$\mathcal{P}(\Gamma_5 y)$, lifts of $\tilde{A}_R^\dagger$	13	85				
$\mathcal{P}(\Gamma_5 y)$, lifts of D	14	50	15	60		
$\mathcal{P}(y \cup \Gamma_5 y)$, lifts of $\tilde{A}_R^\dagger$	11	16				
$\mathcal{P}(y \cup \Gamma_5 y)$, lifts of D	10	9	8	9	OOM	OOM

themselves.) For Wilson fermions, or here for $c = 0$ on the whole lattice, Γ_5 maps right singular vectors of D to left ones, and a basis closed under Γ_5 contains both. With $c \neq 0$ Eq. (11.6) replaces Γ_5 by $(D_-^\dagger)^{-1}\Gamma_5$, which is close to Γ_5 on smooth vectors; on the even sites the factors C_e and X_{PV} of Eq. (11.4) enter as well. This suggests, without proving it, that the coarse space of $\tilde{A}_R^\dagger$ needs the vectors $\Gamma_5 y_j$ in addition to the y_j.

The chirality split does not provide it. The prolongation is already closed under Γ_5, $\Gamma_5\mathcal{P} = \mathcal{P}\Sigma$ (Eq. (6.3)): each lifted vector is split into its Γ_5 components before orthonormalization. This does not put the lifts of $\Gamma_5 y_j$ into $\mathrm{range}\,\mathcal{P}$, because the lifts do not commute with Γ_5: $C_e^{-1}\Gamma_5 y \neq \Gamma_5 C_e^{-1} y$, since $\Gamma_5 C_e \Gamma_5 = C_e^\dagger \neq C_e$. For the same reason a Petrov–Galerkin correction with restriction $\mathcal{P}^\dagger$ and prolongation $\Gamma_5\mathcal{P}$ is the Galerkin one: $\Gamma_5\mathcal{P} = \mathcal{P}\Sigma$ has the range of $\mathcal{P}$. A Petrov–Galerkin correction that differs would need a second set of stored vectors.

Measurements. Table 3 compares prolongations for $\tilde{A}_R^\dagger$ made from the test vectors of one setup for D (32 test vectors, blocks 4^4, the cycle of $\tilde{A}_R$ with the second coarse factor by BiCGStab, GCR(8) to 10^{-12}, one random right-hand side, $m = 0.001$). "Lifts of D" means Eqs. (6.4)–(6.5), "lifts of $\tilde{A}_R^\dagger$" the exchanged lifts above. The overlap of $\mathrm{span}\{\Gamma_5 y_j\}$ with $\mathrm{span}\{y_j\}$ is 10^{-3} of the norm on 12^4 and less on 16^4: the two sets are nearly orthogonal. Half of each $(16 + 16)$ does not converge on 12^4, and a full setup of its own for $\tilde{A}_R^\dagger$ (filter and bootstrap on $\tilde{A}_R\tilde{A}_R^\dagger$) stalls in setup S on 12^4 as well. The union $y \cup \Gamma_5 y$ converges everywhere in about the iterations of $\tilde{A}_R$; the shared $\mathcal{P}(y)$ converges on 16^4 and 24^4 in about twice those and fails only on 12^4, setup S, and there only at $m = 0.001$ (15–18 iterations at $m = 0.01$). That configuration is the one with an unpaired near-zero mode ($Q = +1$, Table 4); whether the stall comes from it, whose left and right singular vectors differ in chirality, has not been checked. The union doubles the test vectors and quadruples the coarse operators, so a solve costs 2–4 times one with the shared $\mathcal{P}$. The shared $\mathcal{P}$ is therefore the default, and the union an option for when the solves with $D^\dagger$ stall (Appendix B).

XII. Numerical results

A. Setups

Numerical setup. The gauge configurations are quenched, generated with the plaquette gauge action at $\beta = 6.0$, which corresponds to a lattice spacing of about 0.1 fm; the volumes are 12^4, 16^4, 24^4 and $24^3 \times 48$, one configuration each. On them two setups of the Möbius operator are used (Table 4), both with $(b, c) = (1.5, 0.5)$, the choice of the JLQCD and the RBC/UKQCD Collaborations, and $L_s = 8$.

- *Setup B (bare links).* The domain-wall height $M_5 = 1.8$ is the value typical of practical simulations on unsmeared links, as adopted by the RBC/UKQCD Collaboration: on rough fields the lattice

TABLE 4. The two setups and the multigrid of the mass scan. All runs: Möbius $(b, c) = (1.5, 0.5)$, $L_s = 8$, $m_{\mathrm{PV}} = 1$, $\alpha_{\mathrm{PV}} = 1$, boundary $(1, 1, 1, -1)$, $\alpha = 0.6$ unless stated. Configurations: quenched, plaquette action at $\beta = 6.0$, on 12^4, 16^4, 24^4 and $24^3 \times 48$ (the 12^4 one has one unpaired near-zero mode, $Q = +1$).

	setup S (stout)	setup B (bare)
M_5	1.0	1.8
gauge smearing	stout, $\rho = 0.1$, 3 steps	none
Chebyshev smoother ν, κ_s	8, 10	12, 30
multigrid (both setups)		
aggregates	4^4 ($24^3 \times 48$: levels below with super-blocks 2, 3)	
test vectors	32 (16 filtered low modes of $\tilde{A}_R^\dagger \tilde{A}_R$, lifted to $\{s, \hat{t}\}$)	
setup	filter 2 rounds; bootstrap 2 rounds of 4 FGMRES steps	
coarse solves	single precision, $\lvert r\rvert^2/\lvert b\rvert^2 < 10^{-2}$	
precision	$\mathcal{P}$ half, cycle single, outer solver double	
outer solver	GCR(2) on $\tilde{A}_L$ to $\lvert r\rvert^2/\lvert b\rvert^2 < 10^{-18}$	

artifact of the Wilson kernel shifts the best value of M_5 above 1.

- *Setup S (stout links).* Three steps of the stout smearing of Sec. IV.E with $\rho = 0.1$ are applied to the links of the fermion operator, the setup adopted by the JLQCD Collaboration. Then $M_5 = 1.0$ is used, since the smearing largely suppresses the artifact that shifts the best value of M_5 away from 1.

The quark mass ranges over $m = 0.001$–0.1, and the Pauli–Villars mass is 1. Timings are from an NVIDIA A30; an RTX 3080 was used for iteration counts and spectral quantities only. The implementation is described in Appendix B.

What the two setups mean for the multigrid. The residual breaking of chiral symmetry at finite L_s comes from the eigenvalues near zero of the Möbius kernel $H_M = \gamma_5(b + c)W\big(2 + (b - c)W\big)^{-1}$ [8], that is from near-zero modes of $W = D_W(-M_5)$, which on rough gauge fields come from small dislocations. The two setups keep them small in two ways, smoothing the field (S) or moving M_5 (B), and so give the multigrid low modes of different character. It shows in three measured places: the smoother needs the interval $[\lambda_{\mathrm{hi}}/10, \lambda_{\mathrm{hi}}]$ in S and $[\lambda_{\mathrm{hi}}/30, \lambda_{\mathrm{hi}}]$ in B, and with B's interval setup S stagnates, with S's setup B needs nearly three times the iterations (Sec. XII.H); the field of values of $\tilde{A}_R$ reaches less far below zero with smearing ($\mathrm{Re}\, W \geq -4.6 \times 10^{-3}$ against -5.0×10^{-2} on 12^4, Sec. III); and the test vectors of $\tilde{A}_R$ are much closer to null vectors of $\tilde{A}_R^\dagger$ in S than in B ($\|\tilde{A}_R^\dagger y\|^2/\|y\|^2 \approx 0.015$ against 0.27 on 12^4 and 16^4, while $\|\tilde{A}_R y\|^2/\|y\|^2 \approx 0.003$ in both), so which prolongation the solve with $D^\dagger$ needs depends on the setup (Sec. XI.C, Table 3).

B. Field of values: the coarse correction amplifies

The next four subsections ask why the method converges although the usual theory does not apply. Two notions from non-Hermitian linear algebra are used. The *field of values* $W(X) = \{z^\dagger X z/z^\dagger z\}$ contains the spectrum, and its leftmost point is the smallest eigenvalue of the Hermitian part $(X + X^\dagger)/2$; GCR is guaranteed to converge at every restart length when W lies in the right half-plane [23]. The *ε-pseudospectrum* is the set of z where $\sigma_{\min}(z - X) < \varepsilon$, i.e. where $\|(z - X)^{-1}\| > 1/\varepsilon$; for a normal X it is the ε-neighborhood of the spectrum, for a non-normal one it can be much larger, and it bounds the convergence of full GMRES [31].

Why the eigenvalues alone do not decide, for readers used to Hermitian matrices: the eigenvectors of a non-normal matrix are not orthogonal, and can be nearly parallel. A vector then decomposes into

TABLE 5. Real part of the field of values $W(X)$ and $\|I - X\|_2$ for $X = \tilde{A}_R S$, $\tilde{A}_R E$, $\tilde{A}_R \mathcal{M}$ (exact coarse solves). $m = 0.05$ on $4^3 \times 8$ (2^4 aggregates, 8 filtered low modes), $m = 0.001$ otherwise, with the multigrid of the mass scan (16 low modes, 32 test vectors, 4^4, Chebyshev of Table 4) but no levels below the coarse lattice. `preconditioned_fov_device -parts`. On 24^4 only $W(\tilde{A}_R)$ is given (Krylov dimensions 10 and 12 agree to six digits); there the field of values of $\tilde{A}_R \mathcal{M}$ did not settle (the two dimensions differ by 2%) and the eigensolvers did not fit in the memory of the A30.

	$\mathrm{Re}\, W(\tilde{A}_R)$	$\|I - \tilde{A}_R S\|$	$\|I - \tilde{A}_R E\|$	$\mathrm{Re}\, W(\tilde{A}_R \mathcal{M})$	$\|I - \tilde{A}_R \mathcal{M}\|$
$4^3 \times 8$, $M_5 = 1$	$[-5.0 \times 10^{-3}, 2.26]$		18.5	$[0.772, 1.222]$	
12^4 S	$[-4.59 \times 10^{-3}, 1.98]$	1.018	1.94×10^4	$[-23.9, 30.8]$	54.8
12^4 B	$[-5.02 \times 10^{-2}, 4.46]$	1.112	2.86×10^3	$[-1.49, 4.17]$	5.64
16^4 S			1.54×10^4	$[-15.6, 27.1]$	
16^4 B			2.5×10^3	$[-1.42, 4.27]$	
24^4 S	$[-1.31 \times 10^{-2}, 1.98]$				
24^4 B	$[-5.08 \times 10^{-2}, 4.48]$				

eigenvectors with large coefficients of opposite sign that cancel; each component decays as its eigenvalue says, but the cancellation is lost along the way, and $\|p(X)z\|$ can grow for a while although every eigenvalue lies in the right half-plane. The field of values sees this ($z^\dagger X z$ can be negative for such a z), and so does the pseudospectrum. How nearly parallel an eigenvector is to the others is measured by the condition number of its eigenvalue, $\kappa = 1/|y^\dagger x|$ for the right eigenvector x and the left eigenvector y normalized to $\|x\| = \|y\| = 1$: $\kappa = 1$ for a normal matrix, and large κ means the eigenvalue moves by $\kappa\varepsilon$ under a perturbation of size ε, which is what widens the pseudospectrum around it (Secs. XII.C, XII.D).

The coarse correction amplifies some vectors by 10^3–10^4, and the field of values of the preconditioned operator reaches into the left half-plane (this subsection), so the bound of Eisenstat, Elman and Schultz [23] for GCR is lost. Its eigenvalues nevertheless stay at $\mathrm{Re}\,\lambda \geq 0.78$ (Sec. XII.C), and its pseudospectra bound the convergence of full GMRES (Sec. XII.D). The truncated GCR used in practice is not covered by that bound: it converges best when the coarse solves are inexact, and with accurate coarse solves it needs more iterations (setup B) or stagnates (setup S; Sec. XII.E).

The Hermitian part of $\tilde{A}_R \mathcal{M}$ was computed with $\mathcal{M}$ made a fixed linear map: both coarse factors solved to $|r|^2/|b|^2 < 10^{-20}$ in double precision with the exact adjoints, and $\mathcal{M}^\dagger$ built with the same $\mathcal{P}$ and the same smoother interval, so that $\langle z, \tilde{A}_R \mathcal{M} w\rangle$ and $\langle \mathcal{M}^\dagger \tilde{A}_R^\dagger z, w\rangle$ agree to 10^{-10}. The cycle was also taken apart: the smoother alone, $S = p(\tilde{A}_R)$, the coarse correction alone, E of Eq. (6.2), and the cycle $\mathcal{M}$ (Table 5).

On $4^3 \times 8$ the field of values of $\tilde{A}_R \mathcal{M}$ lies in the right half-plane; on 12^4 and 16^4 it does not. The field of values of $\tilde{A}_R$ itself hardly moves with the volume: from 12^4 to 24^4 its upper edge moves by less than 1% and its lower edge goes from -4.6×10^{-3} to -1.3×10^{-2} (S) and stays at -5×10^{-2} (B). The smoother is harmless ($\|I - \tilde{A}_R S\| \approx 1$); the factor that amplifies is the coarse correction, $\|I - \tilde{A}_R E\|$ of order 10^3 to 10^4 against 18.5 on $4^3 \times 8$: $\tilde{A}_R E$ is the identity on the vectors the correction is built from (Proposition 1) but amplifies other directions by three to four orders of magnitude, like a strongly oblique projection. The mechanism is the one Eick and Mawhinney [34] describe for a renormalization-group coarse lattice of Möbius fermions: the coarse inverse is large on the low modes of the coarse operator, the interpolated correction is not exactly a low mode of the fine operator, and the part that misses comes back amplified. There the fine operator is $D^\dagger D$, whose spectrum reaches 520 against 0.05 for its lowest thousand modes; it supplies the four orders of magnitude, and a Chebyshev low-pass filter of degree 150–300 is needed. Here $\tilde{A}_R$, with the Pauli–Villars operator dividing out the high modes, is close to the identity on them, and its field of values ends at 2 (S) and 4.5 (B); the large factor is the coarse inverse itself. The post-smoothing, a Chebyshev polynomial of degree 7–11 on the upper part of the spectrum,

TABLE 6. Eigenvalues of $\tilde{A}\mathcal{M}$ (exact coarse solves) by Krylov–Schur: the smallest real part, the condition numbers $\kappa = 1/|y^\dagger x|$ of the six eigenvalues of smallest real part (left eigenvectors from $\mathcal{M}^\dagger\tilde{A}^\dagger$), the eigenvalues farthest from 1 with their κ, and the radius about 1 of the disc that holds all the others. $m = 0.001$. `preconditioned_fov_device -arnoldi -far 1,0` (`-left` for $\tilde{A}_L$).

	$\min \mathrm{Re}\,\lambda$	κ (6 lowest)	outliers (κ)	others: $\lvert\lambda - 1\rvert \leq$
$4^3 \times 8$, $m = 0.05$, R	0.857	2–4		
$4^3 \times 8$, $m = 0.05$, L	0.854	2.8		0.156 (Ritz)
12^4 S, R	0.868	6.3–14.5	$5.86 + 0.86i$ (11.2)	0.434
12^4 S, L	0.861	7.2–26.2	$5.40 + 0.89i$ (13.9)	0.455
12^4 B, R	0.831	9.4–20.3		0.753
12^4 B, L	0.830			0.747
16^4 S, R	0.794	6.2–12.8	$10.63 + 0.19i$ (4.38)	0.474
			$3.13 + 0.02i$ (4.74)	
16^4 B, R	0.783	13.5–42.9		0.840

plays the role of the filter, and the cycle as a whole keeps $\|I - \tilde{A}_R\mathcal{M}\|$ at 55 (S) and 5.6 (B); how much of the remaining amplification the post-smoothing removes has not been separated. The bound of Ref. [23] for GCR therefore does not apply (Sec. III). Replacing E by the Galerkin correction of $\tilde{A}_R$ itself, $\mathcal{P}(\mathcal{P}^\dagger\tilde{A}_R\mathcal{P})^{-1}\mathcal{P}^\dagger$ (built densely), brings the field of values of setup B into the right half-plane, $[0.121, 1.99]$, and reduces $\|I - \tilde{A}_R\mathcal{M}\|$ to 1.84 (setup S: $[-3.62, 13.1]$, 16.8); but GCR(8) to 10^{-16} then needs more iterations, 16 against 13 (B) and 12 against 10 (S). The field of values is not what decides the iteration count.

C. Spectrum: far from the origin

Although the field of values reaches far into the left half-plane, every eigenvalue lies at $\mathrm{Re}\,\lambda \geq 0.78$ (Table 6), with moderate condition numbers. Setup S has one (12^4) or two (16^4) isolated outliers of large modulus that grow with the volume; all other eigenvalues lie within 0.43–0.47 of 1. The left and right forms have the same spectrum to the precision shown. At $m = 0.005$ on 16^4 the lowest real parts are 0.802 (S) and 0.790 (B).

D. Pseudospectra and a bound for GMRES

For a non-normal operator the eigenvalues alone do not bound the convergence of GMRES; the ε-pseudospectrum, the set where $\sigma_{\min}(z - X) < \varepsilon$, does [15, 31]. If a contour Γ of length L encloses the spectrum of X but not the origin and $\sigma_{\min}(z - X) \geq \varepsilon$ on it, then for any polynomial p_k with $p_k(0) = 1$,

$$\frac{\|r_k\|}{\|r_0\|} \leq \|p_k(X)\| \leq \frac{L}{2\pi\varepsilon} \max_{z\in\Gamma} |p_k(z)|. \tag{12.1}$$

An isolated outlier λ_o outside Γ is removed by the factor $(1 - z/\lambda_o)$ of p_k: then $p_k(X) = p_k(X)(I - P_o)$ with P_o the spectral projector, $\|(z - X)^{-1}(I - P_o)\| \leq \kappa_o/\varepsilon$, and with a circle $|z - c| = r$ and $p_k(z) = (1 - z/\lambda_o)(1 - z/c)^{k-1}$,

$$\frac{\|r_k\|}{\|r_0\|} \leq \kappa_o \frac{r}{\varepsilon} \frac{|\lambda_o - c| + r}{|\lambda_o|} \Big(\frac{r}{c}\Big)^{k-1}. \tag{12.2}$$

On 16^4 setup S has two outliers; the factor κ_o is then replaced by $\|I - P_1 - P_2\| \leq 1 + \kappa_1 + \kappa_2$ and there are two factors of the form $(|\lambda_o - c| + r)/|\lambda_o|$. On the real axis between the origin and the spectrum, $\sigma_{\min}(z - \tilde{A}_R\mathcal{M}) \geq 0.024$ (S) and ≥ 0.031 (B) on 12^4 ($z = -0.5 \ldots 0.7$), and 0.107 (S), 0.271 (B) at the origin.

TABLE 7. $\min_\Gamma \sigma_{\min}(z - \tilde{A}_R\mathcal{M})$ on circles $|z - c| = r$ (16 points each, or the number given), and the bound (12.1) or (12.2): the factor per iteration r/c, the constant, and the iterations it guarantees for $\|r_k\|/\|r_0\| < 10^{-9}$. For the dense circles ε is the lower bound $\min_i \sigma_{\min}(z_i) - h/2$ on the whole circle, h the spacing of the points ($\sigma_{\min}$ is 1-Lipschitz in z); otherwise the smallest sampled value. Lanczos to 10^{-3} relative in σ^2. `preconditioned_fov_device -pseudo`.

	c	r	$\min \sigma_{\min}$	r/c	constant	k for 10^{-9}
12^4 S	1	0.6	0.0285	0.6	220	53
12^4 S, 160 points	1	0.6	0.0281, ≥ 0.0164	0.6	380	54
12^4 S	1	0.5	0.0107	0.5	480	40
12^4 B	1.4	0.93	0.0985	0.66	9.4	57
12^4 B, 96 points	1.4	0.93	0.0980, ≥ 0.0676	0.66	13.8	58
12^4 B	1.3	0.87	0.0860	0.67	10.1	58
16^4 S	1	0.6	0.0262	0.6	190	53
16^4 S	1	0.55	0.0135	0.55	340	47
16^4 B	1.4	1.05	0.1295	0.75	8.1	80

The bound guarantees convergence at a fixed rate, r/c per iteration, without stagnation; it is pessimistic by a factor of 3–5 in iterations (GCR needs 10–17 with exact coarse solves, Sec. XII.E), and its rate is that measured on $24^3 \times 48$, where the median factor per GCR step of $|r|^2$ is 0.23–0.34 (Table 9), i.e. 0.48–0.58 in $|r|$, against $r/c = 0.5$–0.6 for setup S. Setup B is not worse between 12^4 and 16^4 ($\min \sigma$ grows from 0.098 to 0.13); in setup S the outliers grow but their condition numbers fall and ε stays the same.

What is not yet rigorous. (i) With 16 points per circle the spacing is larger than ε, and the sampled minimum says nothing about the points in between. On the two dense circles of 12^4 the spacing is below ε, and since $\sigma_{\min}(z - X)$ is 1-Lipschitz in z the bound holds on the whole circle, $\sigma_{\min} \geq 0.0164$ (S) and ≥ 0.0676 (B); the guaranteed iterations change from 53 to 54 and from 57 to 58. The circles of 16^4 have only the 16 points. (ii) Lanczos approaches $\sigma^2_{\min}$ from above; a rigorous lower bound needs an inclusion theorem (Kato–Temple) with a bound on the second singular value. (iii) That the disc holds all eigenvalues but the outliers rests on the four eigenvalues farthest from c found by Arnoldi. (iv) The bound is for full GMRES with a fixed linear $\mathcal{M}$ (Sec. XII.E). (v) The spectra, and the FOV of $\tilde{A}_R\mathcal{M}$, are not available on 24^4 and $24^3 \times 48$: the eigensolvers do not fit in the memory of the A30. This is a limit of the hardware used here, not of the method.

E. Truncated GCR: inexact coarse solves

With the usual coarse solves to 10^{-2}, GCR(2) needs one to five iterations more than the full method (Table 8). With accurate coarse solves, where $\mathcal{M}$ is close to the fixed map of Secs. XII.B–XII.D, GCR(2) stagnates ($|r|^2 \approx 3 \times 10^{-8}$ after 1000 iterations) while GCR(8) converges in 26 and GCR(≥ 16) in 17. This is what the analysis predicts: the pseudospectra guarantee the full method; a truncated or restarted one has no such guarantee when the field of values contains the origin [23, 32]. That the truncated solver works with inexact coarse solves is an empirical fact, and the obvious explanation does not hold: a coarse solve stopped at 10^{-2} does not leave out the directions in which E amplifies. On the vector x at the left end of the field of values of $\tilde{A}_R\mathcal{M}$ (exact coarse solves; 12^4, `preconditioned_fov_device -parts`), the cycle with coarse solves to 10^{-2} gives $\mathrm{Re}\, x^\dagger \tilde{A}_R\mathcal{M}x = -25.2$ (S) and -1.43 (B), against -24.3 and -1.49 with exact ones, and $\|(I - \tilde{A}_R E)x\| = 1.16 \times 10^3$ and 1.03×10^2, against 1.13×10^3 and 1.12×10^2; with tolerances 10^{-8} and 10^{-4} they are those of the exact solves to within 1%. What the tolerance changes is that the cycle is no longer a fixed linear map: with Krylov coarse solves stopped early

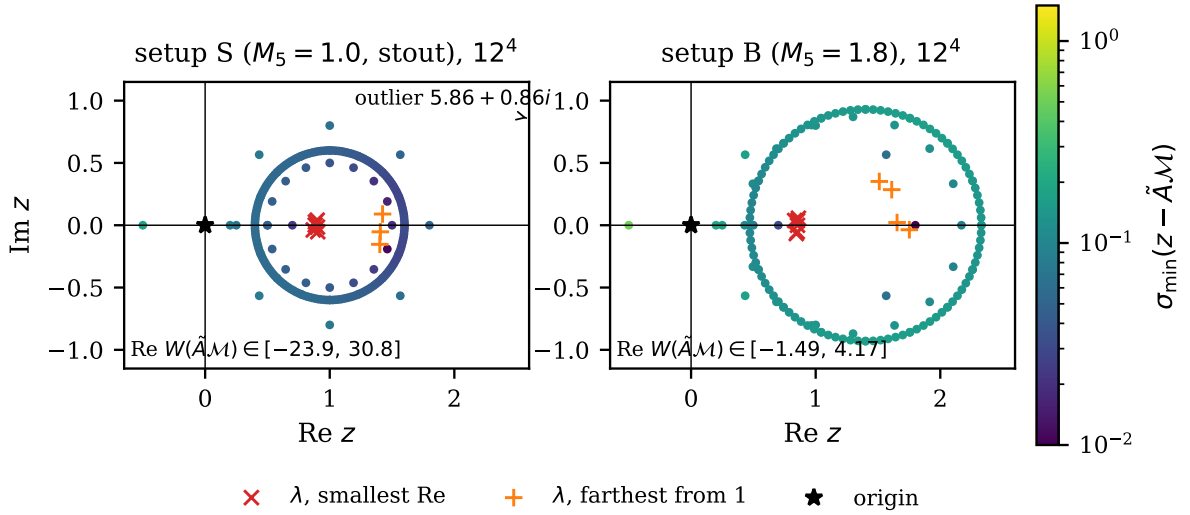


FIG. 5. 12^4, $m = 0.001$, exact coarse solves: the six eigenvalues of $\tilde{A}_R\mathcal{M}$ of smallest real part, the four farthest from 1, and $\sigma_{\min}(z - \tilde{A}_R\mathcal{M})$ (color) at the points where it was computed: the real axis, the contours of Table 7 and the dense circles $|z - 1| = 0.6$ (S) and $|z - 1.4| = 0.93$ (B). The field of values reaches far into the left half-plane (its real range in the corner), the spectrum does not. `data/pseudo_*.tsv`, `data/spectrum_*.tsv`.

TABLE 8. GCR(k) iterations to $|r|^2/|b|^2 < 10^{-18}$, 12^4, $m = 0.001$, the multigrid of Table 4 (no levels below the coarse lattice), one random right-hand side; the point source: range over its 12 columns.

	coarse solves	$k = 2$	8	16	32	64
S, $\tilde{A}_L$	10^{-2}	14			12	
S, $\tilde{A}_R$	10^{-2}	12			11	
B, $\tilde{A}_L$	10^{-2}	14			13	
B, $\tilde{A}_R$	10^{-2}	15			14	
S, $\tilde{A}_L$, point source	10^{-2}	15–19			13–14	
S, $\tilde{A}_L$	10^{-8}	no conv.	26	17	17	17
S, $\tilde{A}_L$	10^{-4}	18			16	

it depends on its input. On $24^3 \times 48$ the full method cannot be run on the A30: GCR(8) and GCR(16) run out of device memory after the setup (GCR(2) peaks at 19.9 of 24 GB, and every further direction is another fine-grid vector).

No run shows a plateau (Table 9, Fig. 6); the first step reduces $|r|^2$ by about 10^{-5} in every run.

What this establishes. As is usual for multigrid preconditioners of lattice Dirac operators, the convergence of the truncated flexible outer method is established numerically: GCR(2) with the coarse solves stopped at 10^{-2} converged in every run of this section, without plateaus, at the rate that the pseudospectral bound of Sec. XII.D gives for the full method. A guarantee for a truncated method whose preconditioner varies from step to step is not available for flexible Krylov methods with multigrid preconditioners in general. That accurate coarse solves make GCR(2) slower, and in setup S stagnate, is an observation of its own; since the loose solves do not avoid the amplifying directions, the variation of the cycle with its input is the candidate, which a fixed but inaccurate linear coarse solve, or the exact cycle perturbed at random, would test.

TABLE 9. Residual history of GCR(2), $24^3 \times 48$, $\tilde{A}_L$, the multigrid of Table 4: steps, median and largest factor of $|r|^2$ per step, and the longest stretch of steps with a factor above $1/2$. From an earlier build than the mass scan; the steps differ from those of Table 10 (`fbc8f00`) by at most one.

setup	m, α	steps	median	worst	slow stretch
S	0.001, 0.6	28	0.315	0.564	2 steps
S	0.001, 1.0	27	0.338	0.479	none
S	0.002, 0.6	23	0.233	0.426	none
S	0.002, 1.0	23	0.253	0.354	none
B	0.001, 0.6	26	0.299	0.357	none
B	0.001, 1.0	28	0.317	0.403	none
B	0.002, 0.6	23	0.246	0.300	none
B	0.002, 1.0	25	0.280	0.347	none

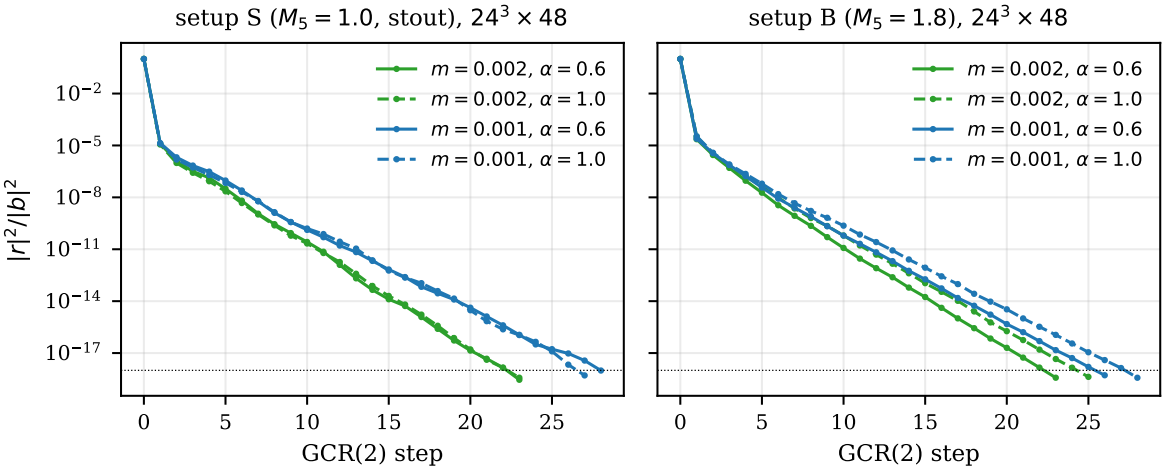


FIG. 6. The GCR(2) residual histories of Table 9, $24^3 \times 48$. `data/gcr_steps_24.tsv`.

F. Comparison with CGNE

What the multigrid costs, against the reference solver of Sec. X, on $24^3 \times 48$: one random right-hand side over seven masses, then whole point-source propagators.

Table 10 and Fig. 7 are the mass scan on $24^3 \times 48$. The CGNE iterations grow by a factor of 21 (S) and 18 (B) from $m = 0.1$ to $m = 0.001$, the multigrid iterations by 2.2 (S) and 3.7 (B). At $m = 0.001$ the multigrid solve is 7.5 times (S) and 9.0 times (B) faster than the double-precision CGNE; with the setup of about 210 s it pays for itself after 2–3 right-hand sides, i.e. within one point-source propagator (12 columns) for $m \leq 0.01$. At $m = 0.1$ in setup S the multigrid is not faster at all. The multigrid needs three times the device memory of CGNE; on the 24 GB A30 this excludes GCR(k) with $k \geq 8$ (out of memory after the setup). Neff's $\alpha = 0.6$ against $\alpha = 1$ in the left form: equal within one iteration for S except at $m = 0.1$ (three fewer), two to three iterations fewer for B. Against an earlier build (`535c9ef`) the iteration counts differ by one or two in six of the 28 multigrid runs; the cause (code change or non-determinism of the device reductions) is not established.

Mixed-precision CGNE. For a comparison on equal terms the reference solver also runs in mixed precision: CG in single precision on the normal equation of $C_e^{-1}S$, with reliable updates in double precision (the true residual recomputed whenever $|r|^2$ has fallen by 10^{-2}). It needs the iterations of double precision (at $m = 0.001$ in setup S 14% more) and 0.54–0.61 times the time, with the same final residuals

TABLE 10. Mass scan on $24^3 \times 48$, A30, one random right-hand side to $|r|^2/|b|^2 < 10^{-18}$ on the even-odd system: even-odd CGNE in double and in mixed precision against the multigrid (GCR(2) on $\tilde{A}_L$), $\alpha = 0.6$ for both and $\alpha = 1$ for the multigrid; iterations, solve time in seconds, coarse-operator applications, and N_{rhs}, the number of right-hand sides at which the multigrid setup (200–237 s) is paid back, $t_{\mathrm{setup}}/(t_{\mathrm{CGNE}} - t_{\mathrm{MG}})$, against the double- and the mixed-precision CGNE. The mixed-precision CGNE needs the iterations of the double-precision one, except at $m = 0.001$ (S): 4090. Final $|D\psi - \eta|^2/|\eta|^2$: 9×10^{-18}–3×10^{-16} (CGNE, both), 2×10^{-19}–3×10^{-18} (MG). Device memory peak: CGNE 6.9 GB (double) and 7.2 GB (mixed), MG 19.9 GB ($\alpha = 0.6$). RutticeQCD `fbc8f00`; `data/massscan_24.tsv`.

	CGNE			MG, $\alpha = 0.6$			MG, $\alpha = 1$		N_{rhs}	
m	it	s	s, mixed	it	s	coarse	it	s	double	mixed
setup S ($M_5 = 1.0$, stout)										
0.1	173	4.8	2.9	13	5.9	1202	16	6.8	—	—
0.05	291	8.1	4.6	13	5.3	750	14	5.7	72	—
0.02	644	17.7	9.7	14	5.7	830	14	5.7	17	51
0.01	1166	32.0	17.4	15	6.2	990	15	6.3	7.9	18
0.005	2067	56.6	30.5	17	7.1	1158	17	7.3	4.2	8.8
0.002	3419	93.5	50.2	23	10.1	1954	22	10.1	2.5	5.1
0.001	3602	98.5	59.9	28	13.2	3014	27	13.1	2.4	4.4
setup B ($M_5 = 1.8$)										
0.1	231	6.4	3.7	7	3.3	201	9	4.2	69	535
0.05	402	11.1	6.2	10	4.7	302	12	5.7	33	143
0.02	857	23.6	12.9	13	6.2	424	15	7.2	12	32
0.01	1447	39.6	21.4	15	7.2	556	17	8.2	6.6	15
0.005	2299	63.0	34.0	18	8.7	714	21	10.2	4.0	8.6
0.002	3473	95.2	51.2	23	11.3	1024	25	12.4	2.8	5.8
0.001	4228	115.6	62.0	26	12.9	1274	28	13.9	2.1	4.3

and 7.2 GB of device memory. Against it the multigrid solve at $m = 0.001$ is 4.5 (S) and 4.8 (B) times faster, and the setup is paid back after 4–5 right-hand sides, within one propagator for $m \leq 0.005$; for $m \geq 0.05$ (S) the mixed-precision CGNE solve is the faster one, and at $m = 0.1$ (B) the two take nearly the same time.

A point-source propagator. The break-even estimates are confirmed by whole propagators, the 12 columns of a point source (Table 11). At $m = 0.001$ the multigrid takes half the wall time of the mixed-precision CGNE including its setup, the solve per column being 4.5–5 times faster; at $m = 0.01$ the setup is not quite paid back within the 12 columns, as $N_{\mathrm{rhs}} \approx 15$–18 (Table 10) predicts. A point source needs more iterations than a random right-hand side, with either solver. The pseudoscalar correlators of the two agree to 4×10^{-8}–7×10^{-7} at $t \leq 3$ and to 2×10^{-6} at all 48 times. The multigrid stops on the residual of D ($|D\psi - \eta|^2/|\eta|^2$ of 1–2 $\times 10^{-18}$), the CGNE on that of its normal equation (6×10^{-17} to 3×10^{-16}). The device memory is 21.8 GB (multigrid) and 9.1 GB (mixed-precision CGNE).

G. Γ_5-Hermiticity for $c \neq 0$

For $c = 0$ the coarse operators are Σ-Hermitian to rounding; for $c \neq 0$ they are not, by a few percent in random directions and up to 14% in the directions of the smallest singular values, growing linearly with c (Table 12), as Eq. (4.10) says. This does not hurt the method: with MINRES on $\mathcal{H}_c$ for the Pauli–Villars factor the outer iterations at $c = 0.5$ are those of the Shamir operator ($c = 0$, a different

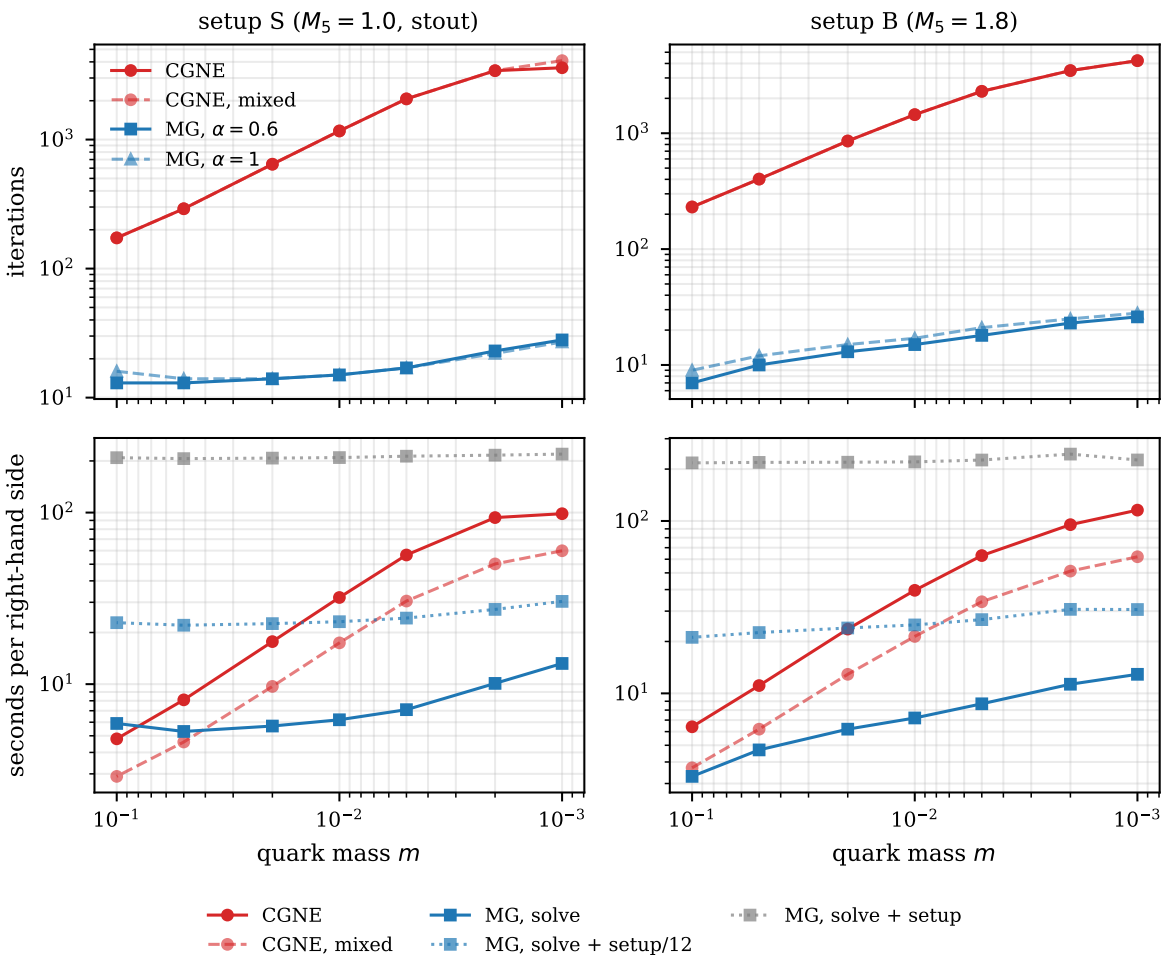


FIG. 7. The mass scan of Table 10 and the mixed-precision CGNE: iterations (top) and seconds per right-hand side (bottom) against the quark mass, $24^3 \times 48$, A30. "Solve + setup/12": the setup spread over the 12 columns of one point-source propagator. `data/massscan_24.tsv`, `figures.py`.

operator, for which MINRES is exact), and MINRES is better than BiCGStab on the exact adjoint in every case (Table 13); with BiCGStab the stout setup nearly stagnates. The non-Hermitian part of $\mathcal{H}_c$ is small enough for MINRES stopped at 10^{-2}; why BiCGStab is so much worse, also at $c = 0$, has not been studied.

H. Sensitivity to the parameters

The method is robust within a window of each parameter and fails outside it (Table 14). The smoother interval must suit the setup; the bootstrap is necessary; more accurate coarse solves are not better, and with GCR(2) the most accurate ones stagnate in setup S (Sec. XII.E). Larger setups (smaller aggregates, more test vectors) buy fewer iterations, but on 12^4 not less time (Table 15). All runs of the table use GCR(2); the stagnations are of the kind of Sec. XII.E, and whether full GCR would remove them has not been tested. Lüscher's test of local coherence for these test vectors has not been made.

Fewer iterations are not always less time (Table 15): 2^4 aggregates need 8–9 iterations but a larger coarse lattice, so the solve and the setup take about twice as long as with 4^4; 48 test vectors save a third of the iterations and lose in setup and in time per iteration. For Wilson fermions, Whyte, Stathopoulos and Romero [35] obtain the effect of more test vectors at the coarse size of fewer: they generate more, truncate them by a singular value decomposition on each aggregate and chirality (as $\mathcal{P}$ is split here by Γ_5), and stream the decomposition to save device memory. Applied here, the truncated basis would no longer

TABLE 11. Point-source propagators (12 columns) on $24^3 \times 48$, A30: the multigrid (setup seconds, iterations and seconds per column) against the mixed-precision CGNE (iterations and seconds per column), the wall time of the whole propagator, and the largest relative difference of the pseudoscalar correlator at $t = 0 \dots 3$. RutticeQCD `fbc8f00`; `data/pointsource_24.tsv`.

		MG			CGNE, mixed		wall (s)		
	m	setup	it	s	it	s	MG	CGNE	$\lvert\Delta C\rvert/C$
S	0.001	206	29–32	13.6–14.9	4182–4572	61.4–67.0	396	810	9×10^{-8}
B	0.001	215	29–30	14.2–14.8	5023–5332	73.5–78.0	409	918	7×10^{-7}
S	0.01	203	16–17	6.4–7.0	1337–1388	19.8–20.5	305	264	2×10^{-7}
B	0.01	215	17	8.1	1641–1745	24.2–25.7	331	324	4×10^{-8}

TABLE 12. $\delta(v) = \|\Sigma X \Sigma v - X^\dagger v\|/\|X^\dagger v\|$ for the coarse operators $X = \mathcal{D}_\mathrm{c} = \mathcal{P}^\dagger D \mathcal{P}$ and $\mathcal{P}^\dagger D_\mathrm{PV} \mathcal{P}$, $X^\dagger$ the exact adjoint, Σ the coarse Γ_5; $b - c = 1$ throughout (as usual for Möbius fermions), $L_s = 8$, $m = 0.001$ ($4^3 \times 8$: $m = 0.05$, 2^4 aggregates, 32 coarse components; 12^4: 4^4, 64). Random v (four) and the right singular vectors of the four smallest singular values. `examples/coarse_hermiticity`. *Lanczos on $X^\dagger X$ not converged in 2000 restarts (the low end of $\mathcal{D}_\mathrm{c}$ is crowded): approximate vectors.

	$\mathcal{P}^\dagger D \mathcal{P}$		$\mathcal{P}^\dagger D_\mathrm{PV} \mathcal{P}$	
	random	low	random	low
$4^3 \times 8$, $M_5 = 1$, $(b, c) = (1, 0)$	3.8×10^{-16}	9×10^{-16}	4.0×10^{-16}	9×10^{-16}
$4^3 \times 8$, $M_5 = 1$, $(b, c) = (1.5, 0.5)$	0.041	0.065–0.074	0.050	0.090–0.095
12^4 S, $(b, c) = (1.25, 0.25)$	0.012	0.034–0.050*	0.030	0.023–0.031
12^4 S, $(b, c) = (1.5, 0.5)$	0.023	0.067–0.099*	0.053	0.041–0.054
12^4 B, $(b, c) = (1.5, 0.5)$	0.030	0.140–0.146	0.055	0.053–0.066

contain s and $\hat{t}$ exactly, so the exactness of Proposition 1 would be lost; whether the gain outweighs that has not been tried. With the setup as fast as it is now, the base setting is no longer the time optimum on 12^4: 16 test vectors (a coarse dimension of 32) halve the setup at a solve that is no slower, and more smoother steps ($\nu = 12$, 16 for setup S) solve faster at the same setup. On $24^3 \times 48$ the fewer test vectors do not carry over (setup S, $m = 0.001$, point source, A30, `fbc8f00`): 16 halve the setup there too (116 against 207 s), but the columns that converge need 910–940 iterations and the others stop at the cap of 2000; 24 (184 s of setup) need 162 iterations for the first column, against 29–32 with 32. More smoother steps have not been tried there. All runs use one setup seed, and the iterations move by a few between builds that add in other orders, so differences of a few iterations are within that variation.

XIII. Open problems

The following questions remain open. Those the measurements answer are discussed where they arise: the left form in Sec. VI.C, $c \neq 0$ in Secs. VII and XII.G, the field of values in Secs. III and XII.B–XII.D.

1. **Choosing the smoother interval.** The interval $[\lambda_\mathrm{hi}/\kappa_s, \lambda_\mathrm{hi}]$ is set from an estimate of the spectral radius and a ratio chosen per setup, and the wrong ratio makes the solve stagnate (Sec. XII.H); the setup works with singular values of $\tilde{A}_R$, which for a non-normal operator need not match its eigenvalues. How to choose the interval from the operator, which part of the low end is left to the smoother after the coarse correction, and whether an ellipse with complex foci should replace the real interval, are open.

TABLE 13. GCR(2) iterations (coarse applications) with the Pauli–Villars factor solved by MINRES on $\mathcal{H}_c$, which assumes $\delta = 0$, or by BiCGStab on the exact adjoint, both to 10^{-2}; 12^4, $m = 0.001$, $\tilde{A}_L$, otherwise Table 4.

	$c = 0.5$, $b = 1.5$		$c = 0$, $b = 1$	
	MINRES	BiCGStab	MINRES	BiCGStab
setup S	14 (3098)	373 (192892)	13 (3061)	24 (7276)
setup B	14 (2351)	19 (3876)	15 (2103)	28 (3540)

TABLE 14. GCR(2) iterations when one parameter of Table 4 is changed, 12^4, $m = 0.001$, $\tilde{A}_L$, one random right-hand side (base: 16 in setup S, 14 in setup B); "stag.": not converged in 1000 iterations. Setup times in seconds where they change much. RTX 3080, RutticeQCD `fbc8f00`; `data/sensitivity_12.tsv`.

change	setup S	setup B
aggregates 2^4 / 3^4 / 6^4	8 (22 s) / 12 / 23	9 (15 s) / 11 / 21
test vectors 8 / 16 / 24 / 48	stag. / 21 / 26 / 10 (17 s)	38 / 18 / 15 / 11 (14 s)
smoother $\nu = 4$ / 6 / 8 / 12 / 16	28 / 18 / 16 / 11 / 9	stag. / 96 / 27 / 14 / 13
interval $\kappa_s = 5$ / 10 / 30 / 100	23 / 16 / stag. / stag.	stag. / 38 / 14 / stag.
coarse solves 10^{-1} / 10^{-2} / 10^{-4} / 10^{-8}	23 / 16 / 18 / stag.	20 / 14 / 16 / 17
bootstrap rounds 0 / 1 / 2 / 3	stag. / 15 / 16 / 22	249 / 46 / 14 / 12
right form $\tilde{A}_R$	11	15

2. **Even-odd smoother and coarse space.** Boyle and Yamaguchi [20] found that a red-black smoother combined with the HDCR coarse correction was not effective, and attributed it to the eigenvectors of the red-black operator and of $\gamma_5 R_5 D$ not being aligned. In the present method the smoother and the correction both approximate $\tilde{A}^{-1}$ and the coarse space is built from low modes of $\tilde{A}_R$, so that particular mismatch is absent by construction; but the coarse operators are Galerkin projections of D and D_{PV}, not of $\tilde{A}$. Whether the even-odd outer system is preferable to the full-lattice A' for this method has not been established.

3. **A rigorous bound, and larger volumes.** The pseudospectral bound rests on Lanczos estimates of $\sigma_{\min}$, which approach it from above, and on the eigenvalues farthest from the centre found by Arnoldi (Sec. XII.D). A bound from an inclusion theorem, and how the outliers of setup S and the pseudospectra develop beyond 16^4, are open. The latter is only a matter of memory: on 24^4 the field of values of $\tilde{A}_R$ fits in the 24 GB of the A30 but the Arnoldi iteration does not. A GPU with more memory would answer it with the programs of Sec. B unchanged; so would the CPU, with enough memory, once the Arnoldi and pseudospectrum options are carried over to the CPU version of the program.

4. **How many test vectors.** Local coherence suggests that the number of test vectors an aggregate needs does not grow with the volume, yet 16 suffice on 12^4 and on $24^3 \times 48$ even 24 do not (Sec. XII.H). Whether the 16 lowest modes of a larger lattice cover too narrow a part of the spectrum, or whether the low modes vary in the fifth dimension more than one vector per aggregate can follow (Eick and Mawhinney [34] find the fifth-dimensional profiles of low modes to differ between lattices), is open; a test vector split in s as it is now split in chirality would keep s and $\hat{t}$ in the range of $\mathcal{P}$, and Lüscher's coherence test would measure what the coarse space captures.

5. **Solving with $D^\dagger$.** The dual cycle used for $\tilde{A}_R^\dagger$ (Sec. X) uses the basis built for D, and the dual correction is not exact on it: for $r = \tilde{A}_R^\dagger y_j$ it misses y_j by 4–90 on $4^3 \times 8$ and by 20–2×10^3 (S)

TABLE 15. The changes of Table 14 that move the cost, timed on the A30: iterations / setup seconds / solve seconds. 12^4, $m = 0.001$, $\tilde{A}_L$, GCR(2), one random right-hand side, RutticeQCD `fbc8f00`; `data/costs_12.tsv`. The iterations are those of Table 14 (RTX 3080, the same build) in every row; with an earlier build they differed by up to six, the device kernels adding in other orders.

change	setup S	setup B
base	16 / 8 / 0.7	14 / 7 / 0.5
aggregates 2^4	8 / 18 / 1.4	9 / 12 / 1.2
aggregates 3^4	12 / 8 / 0.5	11 / 7 / 0.5
aggregates 6^4	23 / 10 / 0.6	21 / 10 / 0.6
test vectors 16	21 / 4 / 0.5	18 / 4 / 0.4
test vectors 24	26 / 6 / 0.9	15 / 6 / 0.4
test vectors 48	10 / 16 / 0.7	11 / 13 / 0.8
smoother $\nu = 4$	28 / 8 / 1.2	stag.
smoother $\nu = 6$	18 / 7 / 0.6	96 / 6 / 3.1
smoother $\nu = 12$	11 / 8 / 0.5	(base)
smoother $\nu = 16$	9 / 8 / 0.4	13 / 7 / 0.5
bootstrap rounds 1	15 / 5 / 0.5	46 / 5 / 1.1
bootstrap rounds 3	22 / 10 / 1.0	12 / 10 / 0.5

and 10^2–2×10^4 (B) on 12^4, more than it misses random vectors (medians 12, 75, 830 against 4.5, 13, 90; exact coarse solves, as in the remarks after Proposition 1). On $4^3 \times 8$ GCR(2) needs about the same iterations with it on $\tilde{A}_R^\dagger$ as with $\mathcal{M}$ on $\tilde{A}_R$ (16 against 15); on 12^4, setup S, $m = 0.001$, it does not converge in 1000 iterations where $\mathcal{M}$ on $\tilde{A}_R$ takes 23 (both coarse factors by BiCGStab to 10^{-2}, double precision, no levels below the coarse lattice). In setup B this program converged with neither cycle, unlike the solver of Sec. XII, so it gives no comparison there. A basis for $D^\dagger$, from the low modes of $\tilde{A}_R \tilde{A}_R^\dagger$ or closed under Γ_5, and how the dual cycle behaves in the solves of HMC, are open.

TABLE 16. The notation of this note (and of RutticeQCD) and that of other references. Equation numbers are those of the references; "the same" means the same operator.

	This note	Ref. [8]	Ref. [24]	Bridge++ code
Height	$M_5 > 0$	$-M_5$, $M_5 = O(-1)$ (2.13)	M_0	M_0, `domain_wall_height`
Fifth dim.	$s = 0, \ldots, L_s - 1$	$s = 1, \ldots, L_s$	$s = 1, \ldots, N_s$	$0, \ldots, N_s - 1$, `extent_of_5th_dimension`
Wilson part	$W = (4 - M_5) - \frac{1}{2}K$	$D^{\mathrm{Wilson}}(M_5)$ (2.13)	$D_{\mathrm{W}}(-M_0)$	$\frac{1}{2\kappa}D_{\mathrm{W}}$, $\kappa = 1/(8 - 2M_0)$
$D_\pm$	$D_+ = 1 + bW$, $D_- = cW - 1$ (Eq. (4.3))	$D_+ = bD^{\mathrm{W}} + 1$, $D_- = cD^{\mathrm{W}} - 1$ (2.18)	the same (2)	—
D	Eq. (4.7)	$D_{\mathrm{DW}}(m)$ (2.20)	$D(m)$ (1)	`Fopr_Domainwall`, mode `D`
C	$d_+ + d_- Y$, Eq. (4.14)	—	$\mathrm{diag}(D_{ee}, D_{oo})$ (4)	D_{ee}, D_{oo} of `_eo`; `m_dp` $= \alpha d_+$, `m_dm` $= \alpha d_-$
Neff's α	DQ, Eq. (4.21)	— (its $\alpha = \frac{b+c}{b-c}$)	—	`parameter_alpha`: DQ, site part CQ
Γ_5	$\gamma_5 R$	$\gamma_5 R D_-^{-1}$ (2.22)	—	—
4d quark	$P_-\psi_0 + P_+\psi_{L_s-1}$	$P_-\Psi_1$, $P_+\Psi_{L_s}$	—	the same

Appendices

A. The notation of other references

Table 16 relates Eqs. (4.6)–(4.21) to Brower, Neff and Orginos [8], to Kanamori, Chen and Matsufuru [24] and to the code of Bridge++ [14] (`Fopr_Domainwall`, `Fopr_Domainwall_eo`, with Neff's α as `parameter_alpha`). Eq. (4.4) is the tridiagonal matrix of Ref. [8], Eq. (2.20), and of Ref. [24], Eq. (1), with $D_+ = 1 + bW$ on the diagonal, $D_- P_+$ below and $D_- P_-$ above it, $-mD_- P_\pm$ in the corners, and s shifted by one. The operators D, C and $D^{(\alpha)}$ are thus the same in all four; the differences are names and signs:

- $D_\pm$ are those of Refs. [8, 24], $D_- = cW - 1$. The hop Y of Eq. (4.6) carries a minus sign, so $D = D_+ - D_- Y$ and $d_- = 1 - c(4 - M_5)$ is minus the site part of D_-. The comments of RutticeQCD on the five-dimensional source call $1 - cW$ "D_-", which is $-D_-$ (the source is that of the Bridge++ domain-wall test, sign included).
- The height $M_5 > 0$ here is M_0 in Ref. [24] and in Bridge++ (`domain_wall_height`); Ref. [8] writes M_5 for the Wilson mass, $M_5^{\mathrm{BNO}} = -M_5$, Eq. (2.13). The caption of Fig. 1 of Ref. [24] gives $M_0 = -1$ for the height 1 of its text.
- The α of Ref. [8], $(b + c)/(b - c)$, is the scale factor of the Möbius kernel (2 at $b = 1.5$, $c = 0.5$), not Neff's α of Eq. (4.21).
- $\Gamma_5 = \gamma_5 R D_-^{-1}$ in Ref. [8], Eq. (2.22); here $\Gamma_5 = \gamma_5 R$.

$D^{(\alpha)}$ here and the operator of Bridge++ with `parameter_alpha` were compared on a 12^4 configuration with stout links, $L_s = 8$, $m = 0.01$: at $\alpha = 0.6$ and $\alpha = 1$ CG on the even-odd normal equation took the same number of iterations for every column of a point source, and the pseudoscalar correlators agree to 10^{-11}. Bridge++ returns $\psi^{(\alpha)}$; $\psi = Q\psi^{(\alpha)}$.

B. Implementation and reproduction

Implementation. The method is implemented in RutticeQCD (Rust), on the CPU and on NVIDIA GPUs through CUDA kernels of the Wilson and Möbius operators taken from Bridge++. On the device the fine operators, the prolongation, the coarse operators and the coarse Krylov solvers all stay on the GPU. The precisions are those of Table 4: the prolongation is stored in half precision unless asked otherwise, the cycle above the coarse lattice runs in single precision on operators converted on the device from the links of the double-precision ones, and the outer solver is in double precision. The coarse solvers are BiCGStab for the quark factor and MINRES on $\mathcal{H}_c$ for the Pauli–Villars factor (Sec. VII), both stopped at $|r|^2/|b|^2 < 10^{-2}$ unless stated, on the even coarse sites. The numbers of Sec. XII come from the parameter-file program `ruttice` and from the examples `preconditioned_fov` (CPU) and `preconditioned_fov_device` (GPU), whose options are given with each result; the remarks after Proposition 1 from the example `correction_exactness` (CPU). The numbers of Sec. XI come from the examples `mobius_gamma5_check` (Eqs. (4.10) and (4.13), CPU), `dminus_cost` (the inverse of D_-, GPU) and `dual_basis` (Table 3, GPU). In HMC the prolongation of the solves with $D^\dagger$ is the shared $\mathcal{P}$ unless `MultiGrid_HMC.dual_with_gamma5` is set, which makes it from $y \cup \Gamma_5 y$. The eigensolvers (thick-restart Lanczos, Krylov–Schur Arnoldi [30] with left eigenvectors, and $\sigma_{\min}(z - X)$ by Lanczos on $(z - X)^\dagger(z - X)$) are in the module `eigen` and are checked against Bridge++'s eigensolver test.

All numbers of Sec. XII come from one configuration per volume; the most useful extensions are more configurations, a larger volume and dynamical ensembles. The rest of this appendix gives what is needed to run them.

Code. RutticeQCD, `https://github.com/atsanroger/RutticeQCD`, branch `cuda-bridge` (the commit of each result is given with it; later commits give the same iterations, up to the two-iteration differences between builds noted in Sec. XII.F). It needs Rust (stable) and, for the GPU, CUDA 12:

```
export CUDA_HOME=/usr/local/cuda
cargo build --release --features cuda-bridge
./target/release/ruttice mg_24.yaml
```

The GPU backend runs on one GPU. The parameter files are in the style of Bridge++ and are documented in `docs/USAGE.md`; gauge configurations are read in the Bridge++ text and binary formats, NERSC, ILDG and SciDAC.

The mass scan (Table 10, Fig. 7). The parameter files are `docs/mg_background/runs/mg_24.yaml` (the multigrid) and `runs/cgne_24.yaml` (the reference, double or mixed precision); the comments in them say how to switch to setup B and to a point source. Each run prints a line

```
seed 91: GCR 28 iterations on the device; 3014 coarse applications;
    worst |Dx-b|^2 2e-18; 13.2 s; ... device 19.91 GB
```

and, for the multigrid, `setup on the device:  206 s`. These, per mass and setup, are what Table 10 lists. On an A30 one configuration takes about 50 minutes for the multigrid (seven masses, both setups, $\alpha = 0.6$; the setup dominates) and under half an hour for the two CGNE scans.

Memory and volume. On $24^3 \times 48$ the multigrid needs 19.9 GB of device memory (CGNE 6.9 GB in double, 7.2 GB mixed), roughly proportional to the volume: $32^3 \times 64$ needs about 63 GB (an 80 GB GPU), $48^3 \times 96$ more than one GPU has. The aggregates, 4^4, must divide the lattice, and each level below the coarse lattice must divide the extent of the level above (`MultiGrid_Level2`, `_Level3`; on $32^3 \times 64$, for instance, super-blocks 2 and 2).

The spectral quantities (Secs. XII.B–XII.D). The example `preconditioned_fov_device` computes, with exact coarse solves, the field of values, the parts of the cycle, the eigenvalues with their condition numbers, and $\sigma_{\min}(z - \tilde{A}\mathcal{M})$ on contours, e.g.

```
cargo run --release --features cuda-bridge \
    --example preconditioned_fov_device -- \
    16 16 binary conf.bin 1.0 0.001 1 --ratio 10 \
    --arnoldi --far 1,0 --pseudo --contour 1,0,0.6,16
```

(arguments: L, T, format, file, M_5, m, stout $0/1$). It fits 16^4 in 24 GB and takes about an hour there; $24^3 \times 48$ does not fit. The CPU examples `preconditioned_fov`, `coarse_hermiticity` and `correction_exactness` need no GPU; the last, e.g.

```
cargo run --release --example correction_exactness -- \
    12 binary conf.bin 1.0 0.001 1 --block 4
```

prints $\|e(r) - y\|/\|y\|$ for the right form, the left form and the dual correction, and the GCR(2) iterations of $\tilde{A}_R$ and $\tilde{A}_R^\dagger$ (the smoother of setup S by default, `-steps 12 -ratio 30` for setup B; the exactness checks take about 12 minutes on 12^4 on a desktop CPU).

What would help most.

1. The mass scan of Table 10 on three to five further $24^3 \times 48$ configurations, for the spread of the iterations and of the speed-up.
2. The same on a larger volume ($32^3 \times 64$ on an 80 GB GPU), for the volume dependence of the multigrid iterations and of the setup.
3. The mass scan on a dynamical ($2 + 1$-flavor) ensemble, e.g. that of Ref. [24], where the spectra of Sec. III were taken.
4. The spectral run above on further 16^4 configurations, for how the outliers of setup S and the pseudospectra vary between configurations.

For each run, the result lines above and the commit of the build are enough; the logs contain everything else.

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